

METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

HAZOP REPORT

Hazard and Operability Study of the R-201 methanol converter and synthesis loop

DOCUMENT NO. : MEOH-V1-SA-HZP-001
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Teaching document generated from the HAZOP tab of the methanol converter simulator. It is not a study of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
A	06-Oct-26	Issued for teaching	A. Soottitantawat			

HAZOP CLOSE-OUT SUMMARY

Project: Methanol Converter Ver 1 · R-201 loop Operator Training Simulator · Document MEOH-V1-SA-HZP-001 rev A · 29 actions, all open at issue. The action party fills "Action taken" and the HAZOP leader changes the status to CLOSED when the close-out sheet (Appendix 3) is signed.

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N2 Circulator C-201 and recycle gas	R01	N2.1, N2.2, N2.4	FT-201B + FSLL-201B: own flow transmitter on the C-201 discharge for the low-circulation trip, so I-2 no longer shares FT-201 with FIC-201.	Instrument		OPEN	Risk 8 Medium
N7 Separator V-201 and crude methanol let-down LV-204	R02	N7.1, N7.2	LT-204B + LSLL-204 + XV-204: low-low level on its own transmitter closing a new let-down shut-off valve (gas blow-by protection); LOPA to set its SIL.	Instrument / Process		OPEN	Risk 10 High
N7 Separator V-201 and crude methanol let-down LV-204	R03	N7.1	PSV-205 on the let-down drum sized for full gas blow-by through LV-204 (and LB-204) at loop pressure.	Process		OPEN	Risk 10 High
N5 Converter R-201	R04	N1.2, N1.4, N2.2, N3.2, N5.1, N5.3, N5.6, N5.7	TSHH 2oo3: three elements per bed exit (TE-214A/B/C ...) voted 2oo3, independent of the indication thermocouples.	Instrument		OPEN	Risk 8 Medium
N1 Make-up gas	R05	N1.8, N1.9, N1.10, N5.3	XV-202B: second make-up shut-off valve in series (1oo2) so one stuck valve does not defeat every trip; proof-test both.	Instrument		OPEN	Risk 8 Medium
N5 Converter R-201	R06	N5.3, N5.9	TAHH-2xx wall: converter shell skin thermocouples with alarm (hot gas reaching the pressure shell).	Instrument / Mechanical		OPEN	Risk 8 Medium
N3/N5 E-202 cold and E-201 hot interchangers with the TV-211 bypass / Converter R-201	R07	N3.4, N5.8	AIT-208 validation: lab check of converter-outlet methanol each shift; alarm on analyzer frozen (no change for a set time).	Operations / Instrument		OPEN	Risk 8 Medium
N8 Purge PV-203, blowdown HV-205 and relief PSV-201	R08	N8.1, N8.2	Deviation alarm between PT-203 (PIC-203) and the PSHH-203 transmitter, so a low-reading PT-203 is seen before the trip point.	Instrument		OPEN	Risk 5 Medium
N2 Circulator C-201 and recycle gas	R09	N2.7	Confirm a non-return valve on the C-201 discharge and the vendor anti-surge / reverse-rotation protection.	Mechanical		OPEN	Risk 6 Medium
N1 Make-up gas	R10	N1.4, N1.5, N1.7	Make-up composition analyzer (CO, CO ₂) with alarm and a written response to a reformer upset (cut FIC-202, raise quench margins).	Instrument / Operations		OPEN	Risk 9 Medium
N6 E-203 final cooler (cooling water)	R11	N2.6, N6.1, N6.2	Low-flow alarm on the cooling water to E-203 (TI-253 alone reacts late).	Instrument		OPEN	Risk 9 Medium
N7/N8 Separator V-201 and crude methanol let-down LV-204 / Purge PV-203, blowdown HV-205 and relief PSV-201	R12	N7.1, N8.3	Lock LB-204 and PB-203 bypass valves closed; bypass use only under permit with an operator stationed at the valve.	Operations		OPEN	Risk 10 High
N6 E-203 final cooler (cooling water)	R13	N6.4, N6.6	Gas detection or high-pressure alarm on the E-203 cooling-water return; confirm CW-side relief for tube rupture.	Process		OPEN	Risk 8 Medium
N4 E-204 start-up steam heater	R14	N4.1, N4.4	Double block and bleed on E-204 steam when idle; steam-side relief sized for tube rupture; steam-flow alarm when HIC-214 is at 0 %.	Process / Instrument		OPEN	Risk 8 Medium
N8 Purge PV-203, blowdown HV-205 and relief PSV-201	R15	N2.8, N3.5, N8.6	Make HV-205 a remotely operated blowdown valve (fail open on ESD or a separate button) with a depressuring rate per API 521 for fire.	Process / Instrument		OPEN	Risk 10 High

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N8 Purge PV-203, blowdown HV-205 and relief PSV-201	R16	N1.8, N3.6, N8.1, N8.8	Confirm PSV-201 relief cases (fire, blocked purge with make-up at maximum pressure, E-201/E-204 tube rupture) and car-seal the relief isolation valves.	Process		OPEN	Risk 5 Medium
N3 E-202 cold and E-201 hot interchangers with the TV-211 bypass	R17	N3.5	Flange management for hot H ₂ service (controlled bolting, hot bolt-up); H ₂ /CO gas detection and UV/IR flame detection at E-201 and the converter nozzles.	Mechanical / Safety		OPEN	Risk 10 High
N3/N5 E-202 cold and E-201 hot interchangers with the TV-211 bypass / Converter R-201	R18	N3.8, N5.9	Verify hot-section materials against the current API 941 (Nelson curves) at the trip temperature plus margin; HTHA inspection programme.	Materials		OPEN	Risk 5 Medium
N7/N9 Separator V-201 and crude methanol let-down LV-204 / Procedures	R19	N7.6, N7.8, N9.9	Closed-loop sampling for make-up, loop gas and crude methanol; CO detectors in the analyzer house; personal CO monitors.	Operations / Safety		OPEN	Risk 8 Medium
N9 Procedures	R20	N4.3, N9.1	Catalyst reduction procedure with hold points (H ₂ in N ₂ steps, bed ΔT limit, water make) and simulator briefing before each reduction.	Operations		OPEN	Risk 9 Medium
N9 Procedures	R21	N9.6, N9.7	Catalyst unloading and entry procedure: passivation end-point or unloading under N ₂ ; inert-entry rules (supplied air, standby, rescue plan); positive isolation.	Operations / Safety		OPEN	Risk 8 Medium
N9 Procedures	R22	N1.2, N4.3, N4.5, N9.2, N9.3	Start-up procedure: admit make-up in MAN at ≤ 25 %, raise slowly while watching bed 1-2 exits; close E-204 steam and its block valve once self-sustaining; train on the simulator.	Operations		OPEN	Risk 9 Medium
N9 Procedures	R23	N9.4	Shutdown procedure: make-up off first, then reduce circulation; never stop C-201 with make-up on.	Operations		OPEN	Risk 4 Low
N9 Procedures	R24	N9.2, N9.8	SIS bypass management: permit, time limit, DCS indication of every bypass, removal checked at shift handover.	Operations		OPEN	Risk 8 Medium
N2/N7 Circulator C-201 and recycle gas / Separator V-201 and crude methanol let-down LV-204	R25	N2.5, N2.8, N7.3, N7.5	Demister ΔP indication on V-201 and a vibration trip on C-201 (liquid carry-over).	Instrument / Mechanical		OPEN	Risk 8 Medium
N5 Converter R-201	R26	N3.1, N3.2, N5.1, N5.2	Position feedback (ZT) on TV-211 and TV-221/231/241 with an output-vs-position deviation alarm (stuck or failed quench valve).	Instrument		OPEN	Risk 9 Medium
N1 Make-up gas	R27	N1.11	Verify on site that every valve goes to its fail position on instrument-air loss; keep XV-202 de-energise-to-trip.	Instrument		OPEN	Risk 6 Medium
N9 Procedures	R28	N1.1, N5.4, N9.5	Restart procedure after a trip: find the cause, reset only when the trip condition has cleared, reheat through E-204, then admit make-up per R22.	Operations		OPEN	Risk 9 Medium
N9 Procedures	R29	N9.6	Nitrogen utility connections by removable spool or double block and bleed with check valve (no syngas backflow into the N ₂ header).	Process		OPEN	Risk 5 Medium

HAZOP REPORT

Hazard and Operability Study of the R-201 methanol converter and synthesis loop
Methanol Converter Ver 1 · R-201 loop Operator Training Simulator

	Role	Name	Signature	Date
Prepared by	HAZOP scribe			
Reviewed by	HAZOP leader (chair)			
Approved by	Plant / project manager			

This report records a teaching HAZOP of the methanol synthesis loop drawn in the simulator. Names, signatures and dates are filled in by the class team that runs the study; the risk matrix is an example for teaching, not a corporate matrix.

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Executive summary

A HAZOP of the R-201 methanol synthesis loop — the make-up gas line with XV-202 and FV-202, the circulator C-201, the E-202 and E-201 interchangers with the TV-211 bypass, the E-204 start-up steam heater, the four-bed quench converter R-201, the E-203 final cooler, the separator V-201 with the crude-methanol let-down LV-204, the purge PV-203 with the blowdown HV-205 and the relief PSV-201, and the procedures for catalyst reduction, start-up, shutdown, restart and maintenance — was carried out with the IEC 61882 guide-word method. The loop was divided into **9 nodes**; **74 deviations** were recorded and **29 recommendations** raised.

Nodes	Deviations	Recommendations	Very high	High	Medium	Low
9	74	29	0	4	58	12

Risk is ranked with the existing safeguards credited (Section 3.4). No deviation was ranked very high; the high-ranked deviations are those where an instrumented trip or the relief valve is the main protection (gas blow-by through LV-204, loss of containment of hot H₂/CO, bed temperature runaway), and they are carried to the LOPA, the SRS of the SIFs (MEOH-V1-IN-SRS-001) and the cause and effect diagram (MEOH-V1-IN-CE-001).

Key findings

- **N7 LV-204 failing open:** the only warning is LIC-204 LO at 02:50; the boot is empty at 05:31 and nothing trips (no SIS function looks at low level). In the plant, gas at 75 bar g would then flow to the let-down drum. The operator has minutes, not hours, which is why LT-204B + LSLL-204 + XV-204 (R02) and PSV-205 (R03) were raised. The model stops the liquid flow when the boot is empty; it does not model the gas blow-by itself.
- **N7 LV-204 stuck closed:** LIC-204 HI at 02:50, then LSHH-204 on its own transmitter trips at 04:36 (level 92.2 %), closing XV-202 and stopping C-201 before liquid reaches the circulator. The high side is protected independently; the low side (above) is not.
- **N8 purge blocked:** PIC-203 HI at 00:40; PSHH-203 trips at 02:54 with a peak of 85.05 bar g — below PSV-201 (88 bar g), so the SIS acts before the relief.
- **N8 PT-203 reading low:** the DCS shows PIC-203 LO (01:19) while the true pressure climbs to 83.0 bar g (above 80 bar g at 03:08); nothing trips because it stays below the PSHH-203 setting of 85 bar g. The alarm points the operator the wrong way — R08 adds a PT-203 vs PSHH-203 deviation alarm.
- **N5 no quench to bed 2:** TIC-221 HI at 00:03 and TI-224 HI at 01:41; bed 2 exit reaches 295.6 °C, 4.4 °C short of the TSHH setting (300 °C), so it runs hot without a trip. The alarms work; the catalyst still spends time well above its normal exit temperature.
- **N2 C-201 trip:** FSLL-201 closes XV-202 at 00:17 with no alarm first; bed 1 exit peaks at 263.4 °C. The trip works, but it reads the same FT-201 as FIC-201 (R01).
- **N1 CO-rich make-up:** production rises to 128.2 t/h and bed 2 exit to 289.7 °C (TI-224 HI at 05:33), no trip. Only the bed alarms tell the operator; there is no make-up composition analyzer (R10).
- **N6 cooling water cut:** TI-253 HI at 00:48; the separator reaches 89.3 °C and production falls to 101.8 t/h while methanol is recycled; nothing trips. An operability problem more than a safety one, but it hides behind a slow alarm (R11).
- **Instrument air failure:** the fail positions take the loop to a safe, dead state: make-up stops, quench and TV-211 open, the beds cool (bed 1 inlet down to 50 °C), pressure falls to 61.5 bar g and the level rises to 58.3 % with LV-204 shut — no trip. Recovery is a full restart (R27, R28).
- **N1 make-up lost:** no trip; circulation carries on and the beds cool (TI-214 down to 85 °C in 30 min), so the restart window is limited — a reason to write the restart procedure (R28).

Actions by action party

Action party	Number	Action No.
Instrument	14	R01, R02, R04, R05, R06, R07, R08, R10, R11, R14, R15, R25, R26, R27
Operations	10	R07, R10, R12, R19, R20, R21, R22, R23, R24, R28
Process	7	R02, R03, R13, R14, R15, R16, R29
Mechanical	4	R06, R09, R17, R25
Safety	3	R17, R19, R21
Materials	1	R18

An action shared by two parties (e.g. Instrument / Operations) is counted under both.

1 Introduction

1.1 Purpose

The HAZOP identifies how the methanol synthesis loop can deviate from its design intent, what the deviations lead to, whether the existing safeguards are adequate, and what should be changed. It also finds operability problems: deviations that harm no one but stop the loop, age the catalyst early, put the crude methanol off its usual make or upset the let-down and purge users. The study feeds the safety layers, LOPA and SRS of the SIFs, and the cause and effect diagram MEOH-V1-IN-CE-001.

1.2 The unit

Fresh syngas from the front end passes the SIS valve XV-202 and the flow valve FV-202 (FIC-202) and joins the recycle gas at the suction of the circulator C-201. Recycle the separator gas plus make-up round the loop at 1928 kNm³/h (FIC-201 → C-201 speed ≈ 80 %), raising it from 75 to about 80.6 bar g (6.5 MW). Heat the converter feed from 49 °C to 97 °C (E-202, 36.5 MW) and the bed-1 feed to 254 °C (E-201, 44.2 MW) while cooling the effluent from 268 to 156 °C; TIC-211 holds bed 1 inlet at 220 °C with TV-211 (fail open). Convert the feed over 208 t of Cu/ZnO/Al₂O₃ in four adiabatic beds, inlets 220 / 225 / 230 / 235 °C, exits ≈ 260 / 277 / 270 / 268 °C, releasing 84.6 MW; hold each bed inlet with its fail-open quench valve. Cool and condense the effluent from 156 °C to 40 °C with cooling water 32 → 41.3 °C so that methanol and water leave the loop in V-201. Separate the condensed crude methanol (131.5 t/h) from the loop gas at 75 bar g and 40 °C; hold the boot level at 50 % with LIC-204 → LV-204 (fail closed) and let the liquid down to the let-down drum at about 6 bar g; never let gas through. Hold the loop at 75 bar g (PIC-203 on PT-203) by purging 105.1 kNm³/h to reformer fuel through PV-203 (fail closed), keeping inerts at about 9.9 % CH₄; depressure to flare through HV-205 in an emergency; relieve through PSV-201 at 88 bar g. The SIS (SIS-201) shuts the make-up through XV-202 on high bed temperature (TSHH-2xx), low circulation (FSLL-201), high separator level (LSHH-204, which also stops C-201) or high loop pressure (PSHH-203), and on the ESD push-button HS-200.

1.3 Objectives

- Identify hazards to people, the environment and the plant from deviations of each node.
- Identify operability problems: catalyst life, production, the crude methanol let-down and the purge to reformer fuel.
- Judge the existing safeguards and rank the risk with the example matrix.
- Raise recommendations and pass the high-ranked scenarios to LOPA.
- Check selected deviations dynamically on the simulator model (Section 5.3).

2 Scope

Boundaries: scope from the make-up gas boundary (upstream of XV-202) round the loop to the let-down drum inlet, the purge to reformer fuel and the flare header, including C-201, the SIS (I-1 ... I-5) and the E-204 steam and E-203 cooling-water connections; excluded: the front-end syngas compressor, the let-down drum and distillation, the flare system itself. Objectives, team, risk matrix.

A node is a section of the loop with one clear design intent. Boundaries are placed where the intent changes: the make-up valve admits gas, the circulator raises pressure, the interchangers exchange heat, the converter reacts, the cooler condenses, the separator splits liquid from gas, the purge and let-down valves drop pressure. Procedures get their own node (N9) because catalyst reduction, admitting make-up, shutdown and restart create deviations that steady operation does not.

2.1 Nodes

Node	Description	Design intent	Design conditions	Equipment / instruments
N1	Make-up gas: boundary → XV-202 → FV-202 → C-201 suction tee	Admit 365.6 kNm ³ /h of fresh syngas (H ₂ 73 / CO 16 / CO ₂ 7 mol %) at 86 bar g and 40 °C at the rate set by FIC-202, and shut it off tightly on any SIS trip.	365.6 kNm ³ /h; 86 bar g; FV-202 ≈ 50 %	Make-up line, XV-202 (SIS shut-off), FV-202 (fail closed), suction tee with the recycle FIC-202, AI-205/206/207 (loop), SIS I-1...I-5 acting on XV-202

Node	Description	Design intent	Design conditions	Equipment / instruments
N2	Circulator C-201 and recycle gas	Recycle the separator gas plus make-up round the loop at 1928 kNm ³ /h (FIC-201 → C-201 speed ≈ 80 %), raising it from 75 to about 80.6 bar g (6.5 MW).	1928 kNm ³ /h; suction 40 °C; discharge 48.6 °C	C-201 (variable speed), recycle line from V-201, discharge to E-202 <i>FIC-201 (FT-201), SI-201, JI-201, TI-202; FSLL-201 (I-2) on the same FT-201</i>
N3	E-202 cold and E-201 hot interchangers with the TV-211 bypass	Heat the converter feed from 49 °C to 97 °C (E-202, 36.5 MW) and the bed-1 feed to 254 °C (E-201, 44.2 MW) while cooling the effluent from 268 to 156 °C; TIC-211 holds bed 1 inlet at 220 °C with TV-211 (fail open).	Feed 79.7 bar g; effluent 268 → 206 → 156 °C; TV-211 ≈ 31 %	E-202, E-201, TV-211 bypass of the E-201 cold side, hot piping to and from R-201 <i>TI-202, TI-203, TI-205, TI-251, TI-252, TIC-211</i>
N4	E-204 start-up steam heater	Heat the bed-1 feed with steam (HIC-214, fail closed; steam ≈ 252 °C) during start-up until the beds reach the 200 °C permissive; idle in normal operation (HIC-214 0 %).	Idle at design; process side at 79.7 bar g	E-204, steam valve (HIC-214), steam block valve, condensate trap <i>HIC-214, FI-214, TIC-211, TI-212</i>
N5	Converter R-201: four beds and quench TV-221 / TV-231 / TV-241	Convert the feed over 208 t of Cu/ZnO/Al ₂ O ₃ in four adiabatic beds, inlets 220 / 225 / 230 / 235 °C, exits ≈ 260 / 277 / 270 / 268 °C, releasing 84.6 MW; hold each bed inlet with its fail-open quench valve.	79.7 bar g inlet; quench 97 °C, 1014 kNm ³ /h; hottest catalyst 276.9 °C	R-201 shell, catalyst beds 1-4, quench distributors, TV-221/231/241, bed thermocouples (3 per bed) <i>TIC-221/231/241, TI-212 ... TI-244, PDI-215 ... PDI-245, AI-208, XI-260; TSHH I-1 on the same bed thermocouples</i>
N6	E-203 final cooler (cooling water)	Cool and condense the effluent from 156 °C to 40 °C with cooling water 32 → 41.3 °C so that methanol and water leave the loop in V-201.	75 bar g gas side; CW about 4 bar g (assumed)	E-203 (gas in tubes, CW in shell, assumed), CW supply and return <i>TI-252, TI-253, TI-254</i>
N7	Separator V-201 and crude methanol let-down LV-204	Separate the condensed crude methanol (131.5 t/h) from the loop gas at 75 bar g and 40 °C; hold the boot level at 50 % with LIC-204 → LV-204 (fail closed) and let the liquid down to the let-down drum at about 6 bar g; never let gas through.	Boot ≈ 15 m ³ at 50 %; LV-204 ≈ 50 %	V-201 with demister and boot, LV-204, LB-204 hand bypass, line to the let-down drum <i>LIC-204 (LT-204), LSHH-204 (own transmitter, I-3), TI-253, FI-209</i>
N8	Purge PV-203, blowdown HV-205 and relief PSV-201	Hold the loop at 75 bar g (PIC-203 on PT-203) by purging 105.1 kNm ³ /h to reformer fuel through PV-203 (fail closed), keeping inerts at about 9.9 % CH ₄ ; depressure to flare through HV-205 in an emergency; relieve through PSV-201 at 88 bar g.	PV-203 ≈ 45 %; PSHH-203 85 bar g; PSV-201 88 bar g	PV-203, PB-203 hand bypass, HV-205, PSV-201, flare header <i>PIC-203 (PT-203), PSHH-203 (own transmitter, I-4), FI-203, AI-207</i>
N9	Procedures: catalyst reduction, start-up and admitting make-up, shutdown, restart, maintenance	Reduce fresh catalyst safely; heat with E-204 and admit make-up only when the SIS permissives hold; shut down with make-up off first; restart only after the cause of a trip is found; hand over for maintenance inert, isolated and passivated.	Procedures in the Exercises tab (cold start-up, hot start, shutdown, ESD)	SIS sequence (reset, admit make-up), E-204, operator actions, catalyst handling <i>SIS screen and permissives, HIC-214, FIC-202 (MAN), TI bed profiles</i>

2.2 Node map

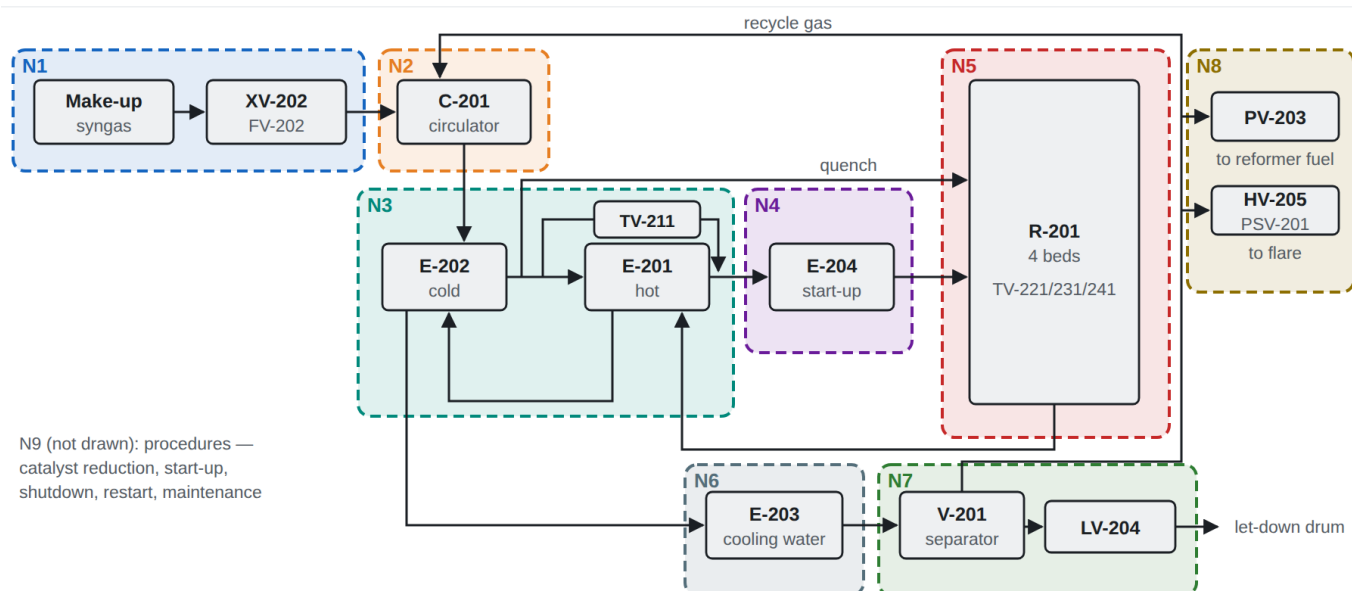


Figure 1. The R-201 loop divided into HAZOP nodes (N9, the procedures node, covers the whole loop).

3 Methodology

HAZOP is a structured, team-based examination of a defined system. Each part of the design is compared with its **design intent**; **guide words** combined with process **parameters** generate **deviations**; for each credible deviation the team records causes, consequences, existing safeguards and, where the risk is not acceptable, recommendations. It finds both hazards (harm to people, environment, assets) and operability problems (the loop cannot make methanol or be run as intended).

3.1 References

Reference	What it contributes
IEC 61882:2016 <i>Hazard and operability studies (HAZOP studies) – Application guide</i>	Definitions, the four phases, guide words, team roles, recording
CCPS <i>Guidelines for Hazard Evaluation Procedures</i> (3rd ed.)	Worksheet format, node selection, "consequence without safeguards", risk ranking, links to LOPA
IEC 61511 (functional safety, SIS)	How HAZOP scenarios feed LOPA and SIL targets for the trips I-1 ... I-4 and the proposed LSSL-204 / XV-204
ISO 31000 / owner's risk matrix	Risk criteria used to decide whether safeguards are adequate
API 521 (pressure relief and depressuring), API 941 (steels for hydrogen service), ASME VIII	Basis for PSV-201 / PSV-205 cases, blowdown rate, HTHA material limits and the pressure envelope
Thailand: Department of Industrial Works regulation on hazard identification, risk assessment and risk-management plans (B.E. 2543)	Legal requirement to identify hazards (HAZOP is an accepted method) and to prepare risk-management plans for factories
OSHA 29 CFR 1910.119 (PSM), where applicable	Process hazard analysis for covered processes (methanol, flammable gas above threshold), revalidated at least every five years

3.2 The four phases

- **Definition:** scope from the make-up gas boundary (upstream of XV-202) round the loop to the let-down drum inlet, the purge to reformer fuel and the flare header, including C-201, the SIS (I-1 ... I-5) and the E-204 steam and E-203 cooling-water connections; excluded: the front-end syngas compressor, the let-down drum and distillation, the flare system itself. Objectives, team, risk matrix.
- **Preparation:** P&ID, PFD-201 (HAZID tab), heat and material balance (Process tab), cause-and-effect chart (SIS screen), alarm list (Workbook tab), procedures (Exercises tab), datasheets; divide into nodes; write each node's design intent; choose parameters and guide words.
- **Examination:** node → design intent → parameter → guide word → is the deviation meaningful? → causes → consequences assuming the safeguards fail → existing safeguards → rank the risk → recommend → next guide word. Record every meaningful deviation, including those with adequate safeguards ("full recording"), so the reasoning can be audited.
- **Documentation and follow-up:** worksheets, recommendation register with owners and dates, close-out, revalidation after changes (MOC) or periodically.

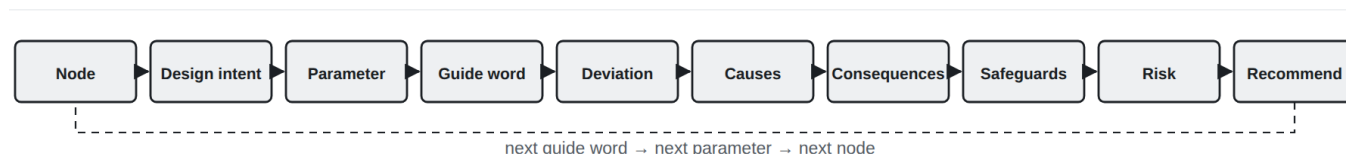


Figure 2. HAZOP examination loop.

3.3 Guide words and parameters

Guide word	Meaning (IEC 61882)	Example in this study
NO / NOT	Complete negation of the intent	Flow no: No make-up gas (N1.1)
MORE	Quantitative increase	Flow more: More make-up than the loop can convert (N1.2)
LESS	Quantitative decrease	Flow less: Less make-up (N1.3)
AS WELL AS	Qualitative increase: something extra present	Composition as well as: Methane slip (more inerts) (N1.6)
PART OF	Qualitative decrease: only part of the intent achieved	Flow part of: Channelling or poor quench mixing (N5.6)
REVERSE	Logical opposite of the intent	Flow reverse: Loop gas back into the front end (N1.9)
OTHER THAN	Complete substitution: something else happens	Composition other than: CO-rich make-up (N1.4)

Guide word	Meaning (IEC 61882)	Example in this study
EARLY	Relative to clock time	Restart early: Restart straight after a trip without finding the cause (N9.5)
LATE	Relative to clock time	Sequence late: Steam left on after the reaction is self-sustaining (N4.5)
BEFORE	Relative to order or sequence	Sequence before: Make-up admitted before the permissives hold (N9.2)
AFTER	Relative to order or sequence	Sequence after: Make-up shut off after the circulator is stopped (shutdown) (N9.4)

Parameters used: flow, pressure, temperature (gas, catalyst and shell), level, composition, heat transfer, integrity (containment), measurement, utility, safety system, material, and for procedures: reduction, admission of make-up, sequence, restart, maintenance, unloading, sampling. A deviation is only recorded if it is physically meaningful ("REVERSE temperature" is not).

3.4 Risk ranking

S \ L	L1	L2	L3	L4	L5
S5	5 (8)	10 (4)	15	20	25
S4	4 (2)	8 (16)	12	16	20
S3	3 (1)	6 (9)	9 (7)	12	15
S2	2 (1)	4 (4)	6 (17)	8 (1)	10
S1	1	2	3 (3)	4 (1)	5

Numbers in brackets: how many deviations of this study fall in the cell. Risk = S × L. 1-4 Low acceptable · 5-9 Medium reduce if practicable (ALARP), review safeguards · 10-16 High recommendation required, go to LOPA · 20-25 Very high not acceptable, act before operation

S	Severity (worst credible consequence)
1	Minor: first aid; small production loss; no release
2	Medical treatment; short outage; small contained release
3	Lost-time injury; loop shutdown for days; catalyst damage; local release
4	Permanent disability or single fatality; major equipment or catalyst loss; fire
5	Catastrophic: multiple fatalities; major fire or explosion; vessel rupture

L	Likelihood with existing safeguards
1	Remote: < 10 ⁻⁴ /y
2	Unlikely: 10 ⁻⁴ - 10 ⁻³ /y
3	Possible: 10 ⁻³ - 10 ⁻² /y
4	Likely: 10 ⁻² - 10 ⁻¹ /y
5	Frequent: > 0.1 /y

Consequences are written assuming the safeguards fail (worst credible case). S ranks that consequence; L ranks its likelihood with the listed safeguards working as designed.

Key scenarios carry the same S × L in the HAZID, HAZOP and Safety layers tabs: gas blow-by through LV-204 S5 × L2 (High, 10); loss of containment of hot H₂/CO/methanol vapour, jet fire S5 × L2 (High, 10); bed temperature runaway (quench valve fails shut, CO-rich gas, loss of circulation with make-up on) S4 × L2 (Medium, 8); CO exposure at sample points S4 × L2 (Medium, 8); pyrophoric reduced catalyst at unloading S4 × L2 (Medium, 8); liquid carry-over into C-201 S3 × L2 (Medium, 6); loop overpressure (purge blocked / PT-203 low) S5 × L1 (Medium, 5).

4 HAZOP sessions

4.1 Schedule

The study is run in class as a teaching HAZOP. A typical plan is three sessions; the team records the dates and attendance on this page.

Session	Nodes	Date	Remark
1	N1 – N4 (make-up gas, circulator, interchangers, start-up heater)		
2	N5 – N8 (converter, final cooler, separator and let-down, purge, blowdown and relief)		
3	N9 (procedures); review of recommendations		

4.2 Team

Role	Responsibility	Name	Attended sessions
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live, reads back recommendations		
Process engineer (designer)	Design intent, balances, bed temperature limits, catalyst behaviour		
Operations representative	How the loop is really run: start-up through the SIS, catalyst reduction, upsets, rounds, sampling		
Instrument / control engineer	Loops, fail positions, SIS logic, which transmitter each trip uses (independence)		
Safety / risk engineer	Risk matrix, LOPA follow-up, relief and depressuring cases		
Specialists as needed	Catalyst vendor (reduction, passivation), materials/inspection (hydrogen service), compressor vendor (C-201), maintenance		

4.3 Documents reviewed

Document	Title / content
MEOH-PID-001 rev A	P&ID sheet 1: R-201 converter, interchanger E-201 and start-up heater E-204 (Appendix 2)
MEOH-PID-002 rev A	P&ID sheet 2: synthesis loop, C-201, E-202, E-203, separator V-201, purge and let-down (Appendix 2)
PFD-201	Process flow diagram with the stream flags (HAZID and PFD tabs)
MEOH-V1-IN-CE-001 rev R0	Cause and effect diagram of the SIS and alarms
MEOH-V1-IN-SRS-001 rev R0	Safety requirements specification of the SIFs
Simulator tabs	Process, Design of the unit operation, Control loop design, alarm and trip list (Workbook tab), start-up and shutdown procedures (SOP tab)

4.4 Nodes reviewed and guide-word coverage

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows
N1	1	2	1	2	·	1	4	·	·	·	·	11
N2	1	2	2	1	·	1	2	·	·	·	·	9
N3	·	3	2	·	·	·	3	·	·	·	·	8
N4	1	2	1	·	·	·	1	·	1	·	·	6
N5	1	4	1	·	1	·	2	·	·	·	·	9
N6	1	2	2	·	·	·	1	·	·	·	·	6
N7	·	2	1	2	·	1	2	·	·	·	·	8
N8	1	3	1	1	·	·	2	·	·	·	·	8
N9	·	2	·	·	·	·	4	1	·	1	1	9

A dot means that guide word gave no meaningful deviation for any parameter in the node (it was still considered). EARLY / LATE / BEFORE / AFTER apply mainly to the procedure node N9.

5 Findings and recommendations

5.1 Risk profile

Node	Deviations	Very high	High	Medium	Low	Recommendations
N1 Make-up gas	11	0	0	10	1	R05, R10, R27
N2 Circulator C-201 and recycle gas	9	0	0	7	2	R01, R09, R25
N3 E-202 cold and E-201 hot interchangers with the TV-211 bypass	8	0	1	5	2	R07, R17, R18
N4 E-204 start-up steam heater	6	0	0	4	2	R14
N5 Converter R-201	9	0	0	9	0	R04, R06, R07, R18, R26
N6 E-203 final cooler (cooling water)	6	0	0	5	1	R11, R13
N7 Separator V-201 and crude methanol let-down LV-204	8	0	2	5	1	R02, R03, R12, R19, R25
N8 Purge PV-203, blowdown HV-205 and relief PSV-201	8	0	1	6	1	R08, R12, R15, R16
N9 Procedures	9	0	0	7	2	R19, R20, R21, R22, R23, R24, R28, R29
Total	74	0	4	58	12	29

High-ranked deviations

Ref	Deviation	Consequence	SxL	Rec.
N3.5	Integrity OTHER THAN: External leak of hot gas at E-201 or the converter nozzles	Jet fire of hot H ₂ /CO/methanol vapour at up to 268 °C with a nearly invisible flame; CO exposure	5x2 = 10	R17, R15
N7.1	Level LESS: Low level: gas blow-by through LV-204	The boot empties (section 9: in a few minutes with LV-204 wide open) and loop gas at 75 bar g flows into the let-down drum at about 6 bar g: drum overpressure and rupture, large release, fire and explosion	5x2 = 10	R02, R03, R12
N7.2	Measurement OTHER THAN: LT-204 freezes	LIC-204 holds a constant output while the real level drifts: overfill (LSHH-204 protects) or empty (nothing independent protects)	5x2 = 10	R02
N8.6	Flow NO: Blowdown not available when needed	Loop stays at pressure in a fire: vessel rupture	5x2 = 10	R15

5.2 Recommendations

No.	Recommendation	Node	Raised from	Action by
R01	FT-201B + FSLL-201B: own flow transmitter on the C-201 discharge for the low-circulation trip, so I-2 no longer shares FT-201 with FIC-201.	N2	N2.1, N2.2, N2.4	Instrument
R02	LT-204B + LSLL-204 + XV-204: low-low level on its own transmitter closing a new let-down shut-off valve (gas blow-by protection); LOPA to set its SIL.	N7	N7.1, N7.2	Instrument / Process
R03	PSV-205 on the let-down drum sized for full gas blow-by through LV-204 (and LB-204) at loop pressure.	N7	N7.1	Process
R04	TSHH 2oo3: three elements per bed exit (TE-214A/B/C ...) voted 2oo3, independent of the indication thermocouples.	N5	N1.2, N1.4, N2.2, N3.2, N5.1, N5.3, N5.6, N5.7	Instrument
R05	XV-202B: second make-up shut-off valve in series (1oo2) so one stuck valve does not defeat every trip; proof-test both.	N1	N1.8, N1.9, N1.10, N5.3	Instrument
R06	TAHH-2xx wall: converter shell skin thermocouples with alarm (hot gas reaching the pressure shell).	N5	N5.3, N5.9	Instrument / Mechanical
R07	AIT-208 validation: lab check of converter-outlet methanol each shift; alarm on analyzer frozen (no change for a set time).	N3/N5	N3.4, N5.8	Operations / Instrument
R08	Deviation alarm between PT-203 (PIC-203) and the PSHH-203 transmitter, so a low-reading PT-203 is seen before the trip point.	N8	N8.1, N8.2	Instrument
R09	Confirm a non-return valve on the C-201 discharge and the vendor anti-surge / reverse-rotation protection.	N2	N2.7	Mechanical
R10	Make-up composition analyzer (CO, CO ₂) with alarm and a written response to a reformer upset (cut FIC-202, raise quench margins).	N1	N1.4, N1.5, N1.7	Instrument / Operations
R11	Low-flow alarm on the cooling water to E-203 (TI-253 alone reacts late).	N6	N2.6, N6.1, N6.2	Instrument

No.	Recommendation	Node	Raised from	Action by
R12	Lock LB-204 and PB-203 bypass valves closed; bypass use only under permit with an operator stationed at the valve.	N7/N8	N7.1, N8.3	Operations
R13	Gas detection or high-pressure alarm on the E-203 cooling-water return; confirm CW-side relief for tube rupture.	N6	N6.4, N6.6	Process
R14	Double block and bleed on E-204 steam when idle; steam-side relief sized for tube rupture; steam-flow alarm when HIC-214 is at 0 %.	N4	N4.1, N4.4	Process / Instrument
R15	Make HV-205 a remotely operated blowdown valve (fail open on ESD or a separate button) with a depressuring rate per API 521 for fire.	N8	N2.8, N3.5, N8.6	Process / Instrument
R16	Confirm PSV-201 relief cases (fire, blocked purge with make-up at maximum pressure, E-201/E-204 tube rupture) and car-seal the relief isolation valves.	N8	N1.8, N3.6, N8.1, N8.8	Process
R17	Flange management for hot H ₂ service (controlled bolting, hot bolt-up); H ₂ /CO gas detection and UV/IR flame detection at E-201 and the converter nozzles.	N3	N3.5	Mechanical / Safety
R18	Verify hot-section materials against the current API 941 (Nelson curves) at the trip temperature plus margin; HTHA inspection programme.	N3/N5	N3.8, N5.9	Materials
R19	Closed-loop sampling for make-up, loop gas and crude methanol; CO detectors in the analyzer house; personal CO monitors.	N7/N9	N7.6, N7.8, N9.9	Operations / Safety
R20	Catalyst reduction procedure with hold points (H ₂ in N ₂ steps, bed ΔT limit, water make) and simulator briefing before each reduction.	N9	N4.3, N9.1	Operations
R21	Catalyst unloading and entry procedure: passivation end-point or unloading under N ₂ ; inert-entry rules (supplied air, standby, rescue plan); positive isolation.	N9	N9.6, N9.7	Operations / Safety
R22	Start-up procedure: admit make-up in MAN at ≤ 25 %, raise slowly while watching bed 1-2 exits; close E-204 steam and its block valve once self-sustaining; train on the simulator.	N9	N1.2, N4.3, N4.5, N9.2, N9.3	Operations
R23	Shutdown procedure: make-up off first, then reduce circulation; never stop C-201 with make-up on.	N9	N9.4	Operations
R24	SIS bypass management: permit, time limit, DCS indication of every bypass, removal checked at shift handover.	N9	N9.2, N9.8	Operations
R25	Demister ΔP indication on V-201 and a vibration trip on C-201 (liquid carry-over).	N2/N7	N2.5, N2.8, N7.3, N7.5	Instrument / Mechanical
R26	Position feedback (ZT) on TV-211 and TV-221/231/241 with an output-vs-position deviation alarm (stuck or failed quench valve).	N5	N3.1, N3.2, N5.1, N5.2	Instrument
R27	Verify on site that every valve goes to its fail position on instrument-air loss; keep XV-202 de-energise-to-trip.	N1	N1.11	Instrument
R28	Restart procedure after a trip: find the cause, reset only when the trip condition has cleared, reheat through E-204, then admit make-up per R22.	N9	N1.1, N5.4, N9.5	Operations
R29	Nitrogen utility connections by removable spool or double block and bleed with check valve (no syngas backflow into the N ₂ header).	N9	N9.6	Process

5.3 Dynamic check of selected deviations on the simulator

A conventional HAZOP judges consequences qualitatively. Here selected deviations are run on copies of the loop model from design conditions with the existing safeguards active (alarms, SIS trips with their 5 s delay, controllers in their normal modes, no operator action) for up to 30 min, or 5 min past a trip. Alarms are evaluated on the model's measurements every second with the page's alarm table and suppression rules. The live session is not touched.

Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trip (first-out)	Range reached (to trip + 5 min, or 30 min)
N1	Flow NO	Make-up gas lost (front-end trip)	FIC-202 LO 00:01 PDI-235 HI 03:26	no trip	loop pressure (true) 58.35 to 75.00 bar g XI-210 0.0 to 112.5 t/h TI-214 85.4 to 260.0 °C
N1	Composition OTHER THAN	CO-rich make-up (reformer upset, 5 min ramp)	TI-224 HI 05:33 TI-214 HI 06:14	no trip	AI-205 CO 4.88 to 6.99 % TI-224 276.9 to 289.7 °C XI-210 112.5 to 128.2 t/h
N2	Flow NO	C-201 trips with make-up on	none in 30 min	FSL-201 circulation below 45 % of design at 00:17	FIC-201 0 to 1928 kNm ³ /h TI-214 260.0 to 263.4 °C loop pressure (true) 68.12 to 75.67 bar g
N5	Flow NO	TV-221 closed (TIC-221 forced to MAN 0 %)	TIC-221 HI 00:03 TI-224 HI 01:41	no trip	TIC-221 225.0 to 258.9 °C TI-224 275.5 to 295.6 °C TI-234 268.5 to 278.0 °C
N6	Flow LESS	Cooling water to E-203 cut by 75 %	TI-253 HI 00:48 PIC-203 HI 01:20 FI-203 HI 01:38 XI-210 LO 02:30 +1 more	no trip	TI-253 40.0 to 89.3 °C loop pressure (true) 73.22 to 78.52 bar g XI-210 101.8 to 112.5 t/h

Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trip (first-out)	Range reached (to trip + 5 min, or 30 min)
N7	Level LESS	LV-204 fails open (LIC-204 forced to MAN 100 %)	LIC-204 LO 02:50 <i>V-201 boot empty (gas path to the let-down drum open): 05:31</i>	no trip	V-201 level (true) 0.0 to 50.0 % FI-209 128.1 to 263.0 t/h loop pressure (true) 75.00 to 75.00 bar g
N7	Level MORE	LV-204 sticks closed	LIC-204 HI 02:50	LSHH-204 separator V-201 level high-high (C-201 also stops) at 04:36	V-201 level (true) 50.0 to 92.2 % loop pressure (true) 67.44 to 75.00 bar g FIC-201 0 to 1928 kNm ³ /h
N8	Pressure MORE	Purge blocked (PIC-203 forced to MAN 0 %)	PIC-203 HI 00:40 FIC-202 LO 01:21 XI-210 LO 02:38	PSHH-203 loop pressure above 85 bar g at 02:54	loop pressure (true) 73.78 to 85.05 bar g AI-207 CH ₄ 9.93 to 13.64 % XI-210 6.4 to 112.5 t/h
N8	Measurement OTHER THAN	PT-203 reads up to 6 bar low (2 min ramp)	PIC-203 LO 01:19 <i>true loop pressure above 80 bar g: 03:08</i>	no trip	loop pressure (true) 75.00 to 82.99 bar g PIC-203 reading 71.32 to 76.99 bar g FI-203 47.6 to 124.0 kNm ³ /h
N1	Utility OTHER THAN	Instrument air failure (all valves to fail position)	FIC-202 LO 00:01 TIC-211 LO 00:03 TIC-221 LO 00:03 TIC-231 LO 00:03 +2 more	no trip	loop pressure (true) 61.48 to 75.00 bar g V-201 level (true) 50.0 to 58.3 % TIC-211 49.9 to 220.0 °C

What the runs tell the HAZOP team

- **N7 LV-204 failing open:** the only warning is LIC-204 LO at 02:50; the boot is empty at 05:31 and nothing trips (no SIS function looks at low level). In the plant, gas at 75 bar g would then flow to the let-down drum. The operator has minutes, not hours, which is why LT-204B + LSL-204 + XV-204 (R02) and PSV-205 (R03) were raised. The model stops the liquid flow when the boot is empty; it does not model the gas blow-by itself.
- **N7 LV-204 stuck closed:** LIC-204 HI at 02:50, then LSHH-204 on its own transmitter trips at 04:36 (level 92.2 %), closing XV-202 and stopping C-201 before liquid reaches the circulator. The high side is protected independently; the low side (above) is not.
- **N8 purge blocked:** PIC-203 HI at 00:40; PSHH-203 trips at 02:54 with a peak of 85.05 bar g — below PSV-201 (88 bar g), so the SIS acts before the relief.
- **N8 PT-203 reading low:** the DCS shows PIC-203 LO (01:19) while the true pressure climbs to 83.0 bar g (above 80 bar g at 03:08); nothing trips because it stays below the PSHH-203 setting of 85 bar g. The alarm points the operator the wrong way — R08 adds a PT-203 vs PSHH-203 deviation alarm.
- **N5 no quench to bed 2:** TIC-221 HI at 00:03 and TI-224 HI at 01:41; bed 2 exit reaches 295.6 °C, 4.4 °C short of the TSHH setting (300 °C), so it runs hot without a trip. The alarms work; the catalyst still spends time well above its normal exit temperature.
- **N2 C-201 trip:** FSLL-201 closes XV-202 at 00:17 with no alarm first; bed 1 exit peaks at 263.4 °C. The trip works, but it reads the same FT-201 as FIC-201 (R01).
- **N1 CO-rich make-up:** production rises to 128.2 t/h and bed 2 exit to 289.7 °C (TI-224 HI at 05:33), no trip. Only the bed alarms tell the operator; there is no make-up composition analyzer (R10).
- **N6 cooling water cut:** TI-253 HI at 00:48; the separator reaches 89.3 °C and production falls to 101.8 t/h while methanol is recycled; nothing trips. An operability problem more than a safety one, but it hides behind a slow alarm (R11).
- **Instrument air failure:** the fail positions take the loop to a safe, dead state: make-up stops, quench and TV-211 open, the beds cool (bed 1 inlet down to 50 °C), pressure falls to 61.5 bar g and the level rises to 58.3 % with LV-204 shut — no trip. Recovery is a full restart (R27, R28).
- **N1 make-up lost:** no trip; circulation carries on and the beds cool (TI-214 down to 85 °C in 30 min), so the restart window is limited — a reason to write the restart procedure (R28).

5.4 Follow-up and close-out

- Each recommendation gets an owner and due date; the owner answers with *accepted and done*, *accepted with alternative* (with justification) or *rejected* (with justification agreed by the HAZOP leader and plant manager).
- High scenarios whose protection relies on instrumented functions (gas blow-by, loss of containment, bed runaway) go to **LOPA** (LOPA tab) to set the required SIL for LSL-204 / XV-204, TSHH 2003 and FSLL-201B (IEC 61511).

- Any change (new catalyst, a higher make-up rate, the agreed additions) goes through **management of change**, which checks whether the HAZOP must be updated.
- Revalidate the HAZOP periodically (typically every five years) and after incidents or near misses.
- Use the simulator to train operators on the scenarios of section 9: the alarm they see first and how much time they have.

Appendix 1 · HAZOP worksheets

Node N1 · Make-up gas

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N1 · Make-up gas: boundary → XV-202 → FV-202 → C-201 suction tee
Design intent	Admit 365.6 kNm ³ /h of fresh syngas (H ₂ 73 / CO 16 / CO ₂ 7 mol %) at 86 bar g and 40 °C at the rate set by FIC-202, and shut it off tightly on any SIS trip.	Design conditions	365.6 kNm ³ /h; 86 bar g; FV-202 ≈ 50 %
Equipment	Make-up line, XV-202 (SIS shut-off), FV-202 (fail closed), suction tee with the recycle	Instruments	FIC-202, AI-205/206/207 (loop), SIS I-1...I-5 acting on XV-202
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	NO	No make-up gas	- Front-end syngas compressor trip - XV-202 closed by a trip or spurious closure - FV-202 fails closed (instrument air)	Production stops; the loop keeps circulating and the beds cool until the reaction dies; loop pressure falls as the purge continues; restart needs the permissives	- FIC-202 L 330 kNm³/h - XI-210 L 105 t/h - PIC-203 closes the purge	2	3	6 M	R28	Operations
N1.2	Flow	MORE	More make-up than the loop can convert	- FIC-202 SP raised (production push) - FT-202 reads low - FV-202 opened in MAN	More heat released in beds 1 and 2 → bed exits towards the catalyst limit; more purge and inerts	- FIC-202 H 400 kNm³/h - TI-214 H 275 °C - TI-224/234/244 H 285 °C - TSHH I-1 at 300 °C closes XV-202	4	2	8 M	R04, R22	Instrument · Operations
N1.3	Flow	LESS	Less make-up	- Front-end rate cut - FV-202 sticking	Lower production; beds cooler; purge falls	- FIC-202 L 330 kNm³/h - XI-210 L 105 t/h	1	3	3 L	—	—
N1.4	Composition	OTHER THAN	CO-rich make-up	Reformer or shift upset: CO up, CO ₂ down	Much more heat per kmol converted in beds 1 and 2: bed exits approach the trip; sintering; by-products	- TIC-211/221/231/241 - TI-214 H 275 °C - TI-224 H 285 °C - TSHH I-1 at 300 °C	4	2	8 M	R10, R04	Instrument · Instrument / Operations
N1.5	Composition	OTHER THAN	CO ₂ -rich make-up	Reformer upset: CO down, CO ₂ up	Less reaction heat, more water in the crude, production falls	- AI-206 indication - XI-210 L 105 t/h	2	3	6 M	R10	Instrument / Operations
N1.6	Composition	AS WELL AS	Methane slip (more inerts)	Reformer methane slip	Inerts build up; lower reactant partial pressure; purge rises; production and carbon efficiency fall	- AI-207 H 13 % - FI-203 H 150 kNm³/h	2	3	6 M	—	—
N1.7	Composition	AS WELL AS	Poisons or metal carbonyls	- Sulfur or chloride slip from the front end - Fe/Ni carbonyls from carbon steel in cold CO service	Top of bed 1 loses activity; heat release moves down; faster ageing; wax and hot spots from Fe/Ni deposits	- Front-end guard beds (outside scope) - bed-1 profile TI-212/213/214	3	3	9 M	R10	Instrument / Operations
N1.8	Pressure	MORE	Loop pressure driven up by the make-up	- Loop not consuming gas (reaction stopped) while make-up flows - front-end compressor at its maximum	Loop pressure rises towards the make-up pressure 86 bar g; flange leaks, rupture if relief fails	- PIC-203 H 78 bar g - PSHH-203 I-4 at 85 bar g closes XV-202 - PSV-201 88 bar g	5	1	5 M	R05, R16	Instrument · Process
N1.9	Flow	REVERSE	Loop gas back into the front end	Front-end pressure falls below the loop with FV-202 open and XV-202 still open	Loop depressures backwards; surge of the front-end compressor	- XV-202 closes only on a loop trip - check valve on the make-up line (assumed)	3	2	6 M	R05	Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.10	Safety system	OTHER THAN	XV-202 fails to close on demand	Stuck valve, solenoid failure: one final element serves every trip	Every trip that relies on stopping the make-up fails: runaway, overpressure or blow-by continue	- FIC-202 is also driven to MAN 0 % by the trip (FV-202 closes) - proof testing	5	1	5 M	R05	Instrument
N1.11	Utility	OTHER THAN	Instrument air failure	Air compressor trip, header break	All valves to fail positions: FV-202, PV-203, LV-204 shut; TV-211 and quench valves open → reaction dies, level rises slowly (see section 9)	- Fail positions (FC/FO) chosen for a safe state - XV-202 de-energise-to-trip	2	3	6 M	R27	Instrument

Node N2 · Circulator C-201 and recycle gas

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N2 · Circulator C-201 and recycle gas
Design intent	Recycle the separator gas plus make-up round the loop at 1928 kNm ³ /h (FIC-201 → C-201 speed ≈ 80 %), raising it from 75 to about 80.6 bar g (6.5 MW).	Design conditions	1928 kNm ³ /h; suction 40 °C; discharge 48.6 °C
Equipment	C-201 (variable speed), recycle line from V-201, discharge to E-202	Instruments	FIC-201 (FT-201), SI-201, JI-201, TI-202; FSLL-201 (I-2) on the same FT-201
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Flow	NO	No circulation	- C-201 motor trip - power failure - LSHH-204 or ESD stops C-201	If make-up continues: gas meets stagnant hot catalyst → hot spot in bed 1, sintering	- FSLL-201 I-2 below 45 % (868 kNm³/h) closes XV-202 — uses FT-201 of FIC-201 - FIC-201 L 1700 kNm³/h	4	2	8 M	R01	Instrument
N2.2	Flow	LESS	Low circulation	- FIC-201 SP lowered - fouled impeller (performance loss) - FT-201 reading high	Less heat carried out of the beds per kmol → bed exits hotter; at low enough flow the TSHH trips	- FIC-201 L 1700 - TI-224/234/244 H 285 °C - TSHH I-1 - FSLL-201 I-2	4	2	8 M	R01, R04	Instrument
N2.3	Flow	MORE	High circulation	- FIC-201 SP raised - FT-201 reading low	Lower per-pass conversion, beds cooler, more power and bed ΔP	- FIC-201 H 2350 - speed limit 100 %	1	3	3 L	—	—
N2.4	Measurement	OTHER THAN	FT-201 fails reading high	Impulse-line or transmitter fault	FIC-201 cuts the real circulation and the FSLL-201 trip reads the same wrong value: loss of circulation not seen by I-2	- Bed temperature alarms - TSHH I-1 (independent of flow)	4	2	8 M	R01	Instrument
N2.5	Composition	AS WELL AS	Liquid in the circulator suction	- V-201 high level - demister damaged - separator hot	Liquid slugs damage impeller and seals → gas leak; circulation lost	- LIC-204 H 75 % - LSHH-204 I-3 at 90 % stops C-201 and closes XV-202 - TI-253 H 45 °C	3	2	6 M	R25	Instrument / Mechanical
N2.6	Temperature	MORE	Hot suction gas	- Cooling-water loss at E-203 - hot afternoon	Higher discharge temperature and power; methanol vapour recycled to the converter	- TI-253 H 45 °C - TI-202 indication	2	3	6 M	R11	Instrument
N2.7	Flow	REVERSE	Reverse flow through C-201 after a stop	C-201 stops while the discharge side is at higher pressure	Machine turns backwards; seal and bearing damage; leak	Discharge non-return valve (vendor, assumed)	3	2	6 M	R09	Mechanical
N2.8	Integrity	OTHER THAN	Seal or casing leak	- Dry-gas seal failure - vibration - liquid slug	Loop gas release at 75 bar g in the compressor house: jet or flash fire, CO exposure	- Seal-gas monitoring - ESD I-5 stops C-201 and closes XV-202 - HV-205	4	2	8 M	R15, R25	Instrument / Mechanical · Process / Instrument
N2.9	Pressure	LESS	Low suction pressure	Loop depressuring (blowdown, purge wide open, make-up lost)	Circulation falls with gas density; FSLL-201 may trip	- FIC-201 raises speed - FSLL-201 I-2	2	2	4 L	—	—

Node N3 · E-202 cold and E-201 hot interchangers with the TV-211 bypass

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N3 · E-202 cold and E-201 hot interchangers with the TV-211 bypass
Design intent	Heat the converter feed from 49 °C to 97 °C (E-202, 36.5 MW) and the bed-1 feed to 254 °C (E-201, 44.2 MW) while cooling the effluent from 268 to 156 °C; TIC-211 holds bed 1 inlet at 220 °C with TV-211 (fail open).	Design conditions	Feed 79.7 bar g; effluent 268 → 206 → 156 °C; TV-211 ≈ 31 %
Equipment	E-202, E-201, TV-211 bypass of the E-201 cold side, hot piping to and from R-201	Instruments	TI-202, TI-203, TI-205, TI-251, TI-252, TIC-211
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Temperature	LESS	Bed 1 inlet too cold	- TV-211 fails open (instrument air) or sticks open - E-201 fouled - converter outlet cooler	Bed 1 slows, the heat front moves down, the reaction can die out; production falls	- TIC-211 L 212 °C - XI-210 L 105 t/h	2	3	6 M	R26	Instrument
N3.2	Temperature	MORE	Bed 1 inlet too hot	- TV-211 sticks closed - TIC-211 thermocouple reading low - E-204 steam passing	Bed 1 exit rises; catalyst sintering; runaway risk	- TIC-211 H 228 °C - TI-214 H 275 °C - TSHH I-1 at 300 °C	4	2	8 M	R04, R26	Instrument
N3.3	Heat transfer	LESS	E-201 fouling	Wax or carbonyl deposits	Less preheat: TV-211 closes to compensate until it saturates, then bed 1 inlet falls; more heat left for E-202/E-203	- TIC-211 - TV-211 output indication	2	3	6 M	—	—
N3.4	Integrity	OTHER THAN	E-201 internal leak	- Tube or tubesheet failure - cold feed at the higher pressure	Feed bypasses the converter: lower conversion and effluent temperature, hard to diagnose	- AI-208 L 3.8 % - XI-210 L 105 t/h	2	3	6 M	R07	Operations / Instrument
N3.5	Integrity	OTHER THAN	External leak of hot gas at E-201 or the converter nozzles	- Thermal cycling at start-up and trips - bolt relaxation - gasket failure	Jet fire of hot H ₂ /CO/methanol vapour at up to 268 °C with a nearly invisible flame; CO exposure	- Gas detection (if fitted) - ESD - HV-205 blowdown (manual)	5	2	10 H	R17, R15	Mechanical / Safety · Process / Instrument
N3.6	Pressure	MORE	Exchanger side blocked in while hot	Isolation for maintenance with heat on the other side	Thermal expansion of trapped gas → overpressure of the isolated side	PSV-201 covers only equipment connected to the loop	3	1	3 L	R16	Process
N3.7	Temperature	MORE	Hot effluent to E-203	- E-202 fouled - converter outlet hotter	More load on E-203; separator hotter; methanol recycled	- TI-252 indication - TI-253 H 45 °C	2	2	4 L	—	—
N3.8	Material	OTHER THAN	Hot-section steel below its hydrogen limit (HTHA)	- Material not per API 941 at H₂ partial pressure ≈ 65 bar a - missing PWHT	Hidden hydrogen attack → sudden rupture, large release, fire	- Material selection - inspection	5	1	5 M	R18	Materials

Node N4 · E-204 start-up steam heater

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N4 · E-204 start-up steam heater
Design intent	Heat the bed-1 feed with steam (HIC-214, fail closed; steam ≈ 252 °C) during start-up until the beds reach the 200 °C permissive; idle in normal operation (HIC-214 0 %).	Design conditions	Idle at design; process side at 79.7 bar g
Equipment	E-204, steam valve (HIC-214), steam block valve, condensate trap	Instruments	HIC-214, FI-214, TIC-211, TI-212
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Flow	MORE	Steam in normal operation (valve passing)	- HIC-214 valve leaking through - block valve left open	Bed 1 inlet hotter; TV-211 opens to compensate and hides it; wasted steam; with TV-211 saturated bed 1 overheats	- TIC-211 H 228 °C - FI-214 indication - TI-214 H 275 °C	2	3	6 M	R14	Process / Instrument
N4.2	Flow	NO	No steam at start-up	- Steam header down - block valve closed - trap blocked	Beds cannot reach the 200 °C permissive; start-up delayed	Start permissive: TIC-211 and TI-212 ≥ 200 °C	1	3	3 L	—	—
N4.3	Temperature	MORE	Too much heating at start-up	- HIC-214 opened quickly - make-up admitted with a hot bed	Beds above target when make-up arrives → sharp exotherm and trip	- Permissive: no bed above 280 °C - TI alarms - TSHH I-1	3	2	6 M	R20, R22	Operations
N4.4	Integrity	OTHER THAN	E-204 tube leak (gas into steam)	Tube failure	Loop gas at 80 bar g enters the steam and condensate system: overpressure, flammable gas at condensate vents	Steam block valve closed in normal operation	4	2	8 M	R14	Process / Instrument
N4.5	Sequence	LATE	Steam left on after the reaction is self-sustaining	HIC-214 or its block valve not closed (the SIS closes the steam only on a trip)	As steam passing; E-201 duty falls, loop runs hotter at the same TIC-211 output	- SIS closes E-204 steam on every trip - procedure	2	3	6 M	R22	Operations
N4.6	Heat transfer	LESS	Condensate backs up in the shell	Trap failure	Low heater duty; water hammer on the next start-up	Operator rounds	2	2	4 L	—	—

Node N5 · Converter R-201

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N5 · Converter R-201: four beds and quench TV-221 / TV-231 / TV-241
Design intent	Convert the feed over 208 t of Cu/ZnO/Al ₂ O ₃ in four adiabatic beds, inlets 220 / 225 / 230 / 235 °C, exits ≈ 260 / 277 / 270 / 268 °C, releasing 84.6 MW; hold each bed inlet with its fail-open quench valve.	Design conditions	79.7 bar g inlet; quench 97 °C, 1014 kNm ³ /h; hottest catalyst 276.9 °C
Equipment	R-201 shell, catalyst beds 1-4, quench distributors, TV-221/231/241, bed thermocouples (3 per bed)	Instruments	TIC-221/231/241, TI-212 ... TI-244, PDI-215 ... PDI-245, AI-208, XI-260; TSHH I-1 on the same bed thermocouples
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Flow	NO	No quench to bed 2	- TV-221 fails closed or sticks - TIC-221 in MAN at 0 % - TIC-221 thermocouple reads high	Bed 2 inlet rises by tens of °C; bed 2 exit approaches the trip; sintering; runaway	- TIC-221 H 232 °C - TI-224 H 285 °C - TSHH I-1 at 300 °C	4	2	8 M	R04, R26	Instrument
N5.2	Flow	MORE	Too much quench (valve fails open)	- TV-231 fails open - instrument air failure (FO) - TIC thermocouple reads low	Next bed too cold; reaction shifts to the other beds, bed 1 and 2 run hotter as the main-feed share falls; production falls	- TIC-231 L 223 °C - TI-214 H 275 °C - XI-210 L 105 t/h	3	3	9 M	R26	Instrument
N5.3	Temperature	MORE	Bed temperature runaway	- CO-rich gas - loss of circulation with make-up on - high make-up - E-204 steam passing	Catalyst above ~300 °C: sintering, by-products (wax, higher alcohols, methane); hot gas at the shell → shell overheating	- TI-214 H 275 °C, TI-224/234/244 H 285 °C - TSHH I-1 at 300 °C (same elements as the indication) - FSLL-201 I-2 - XV-202 closes	4	2	8 M	R04, R05, R06	Instrument · Instrument / Mechanical
N5.4	Temperature	LESS	Beds too cold, reaction dies	- Make-up lost - quench open - E-201 fouled	Production stops; restart needs reheating through E-204	- XI-210 L 105 t/h - TIC L alarms	2	3	6 M	R28	Operations
N5.5	Pressure	MORE	High bed pressure drop	- Catalyst crushing and fines (bed 3) - wax	Lower circulation; maldistribution and hot spots; support-grid damage	- PDI-235 H 1.3 bar - FIC-201	3	3	9 M	—	—
N5.6	Flow	PART OF	Channelling or poor quench mixing	- Damaged quench distributor - catalyst settling	Part of the gas bypasses the catalyst; local hot and cold zones that one thermocouple misses	Three thermocouples per bed (upper third, lower third, exit)	3	2	6 M	R04	Instrument
N5.7	Measurement	OTHER THAN	Bed thermocouple fails low	Open circuit (reads ambient)	That element cannot trip the TSHH; the operator loses the bed-exit reading	- TI-212/213 in the same bed - the other TSHH elements	4	2	8 M	R04	Instrument
N5.8	Composition	OTHER THAN	Catalyst ageing or poisoning	- Sintering over the years - sulfur or chloride slip	Lower activity; inlets raised to compensate; hot zone moves down	- Bed profiles - AI-208 L 3.8 %	2	4	8 M	R07	Operations / Instrument
N5.9	Temperature	MORE	Hot gas at the pressure shell	- Internal leak into the annulus - insulation damage	Shell above its design temperature: HTHA, creep, rupture	None on the P&ID (no skin thermocouples)	5	1	5 M	R06, R18	Instrument / Mechanical · Materials

Node N6 · E-203 final cooler (cooling water)

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N6 · E-203 final cooler (cooling water)
Design intent	Cool and condense the effluent from 156 °C to 40 °C with cooling water 32 → 41.3 °C so that methanol and water leave the loop in V-201.	Design conditions	75 bar g gas side; CW about 4 bar g (assumed)
Equipment	E-203 (gas in tubes, CW in shell, assumed), CW supply and return	Instruments	TI-252, TI-253, TI-254
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Flow	LESS	Less cooling water	- CW pump trip - valve throttled - fouling	Separator hot: methanol stays in the gas and is recycled; production falls; purge and loop pressure rise; hotter C-201 suction	- TI-253 H 45 °C - PIC-203 H 78 - FI-203 H 150 - XI-210 L 105 t/h	3	3	9 M	R11	Instrument
N6.2	Flow	NO	No cooling water	- CW header lost - both valves closed	As above, faster; CW trapped in the tubes may boil	TI-253 H 45 °C	3	2	6 M	R11	Instrument
N6.3	Temperature	MORE	Hot cooling-water supply	- Hot afternoon - cooling-tower problems	Separator a few °C hotter; small production loss	TI-253 H 45 °C	1	4	4 L	—	—
N6.4	Integrity	OTHER THAN	Tube leak: loop gas into the cooling water	- Corrosion - vibration	Loop gas carried to the cooling tower: flammable and toxic gas at the tower; CW overpressure	CW-side relief (assumed)	4	2	8 M	R13	Process
N6.5	Heat transfer	LESS	Fouling on the CW side	Scaling, biofouling	Slow rise of TI-253	- TI-253 H - cleaning programme	2	3	6 M	—	—
N6.6	Pressure	MORE	CW side blocked in with hot gas flowing	CW inlet and outlet valves closed	Boiling and overpressure of the CW side	CW-side relief (assumed)	3	2	6 M	R13	Process

Node N7 · Separator V-201 and crude methanol let-down LV-204

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N7 · Separator V-201 and crude methanol let-down LV-204
Design intent	Separate the condensed crude methanol (131.5 t/h) from the loop gas at 75 bar g and 40 °C; hold the boot level at 50 % with LIC-204 → LV-204 (fail closed) and let the liquid down to the let-down drum at about 6 bar g; never let gas through.	Design conditions	Boot ≈ 15 m³ at 50 %; LV-204 ≈ 50 %
Equipment	V-201 with demister and boot, LV-204, LB-204 hand bypass, line to the let-down drum	Instruments	LIC-204 (LT-204), LSHH-204 (own transmitter, I-3), TI-253, FI-209
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N7.1	Level	LESS	Low level: gas blow-by through LV-204	- LV-204 fails open - LB-204 bypass left open - LT-204 frozen or reading high - LIC-204 in MAN with a high output	The boot empties (section 9: in a few minutes with LV-204 wide open) and loop gas at 75 bar g flows into the let-down drum at about 6 bar g: drum overpressure and rupture, large release, fire and explosion	- LIC-204 L 25 % - LV-204 fail closed	5	2	10 H	R02, R03, R12	Instrument / Process · Operations · Process
N7.2	Measurement	OTHER THAN	LT-204 freezes	Transmitter fault	LIC-204 holds a constant output while the real level drifts: overfill (LSHH-204 protects) or empty (nothing independent protects)	LSHH-204 I-3 on its own transmitter (high side only)	5	2	10 H	R02	Instrument / Process
N7.3	Level	MORE	High level in V-201	- LV-204 sticks closed - LT-204 frozen - let-down drum blocked	Liquid carried into the C-201 suction: impeller and seal damage, gas leak, circulation lost	- LIC-204 H 75 % - LSHH-204 I-3 at 90 % on its own transmitter closes XV-202 and stops C-201	3	2	6 M	R25	Instrument / Mechanical
N7.4	Flow	MORE	Surge of crude to the let-down drum	LV-204 opens on a level swing	Let-down drum level and flash gas rise	Let-down drum controls (outside scope)	2	3	6 M	—	—
N7.5	Composition	AS WELL AS	Liquid carried over with the gas	- Demister damaged - high gas velocity	Methanol recycled to the converter (equilibrium penalty); liquid to C-201	None direct	2	3	6 M	R25	Instrument / Mechanical
N7.6	Composition	AS WELL AS	Dissolved gas in the crude	H ₂ , CO, CO ₂ dissolved at 75 bar g	Flash gas at the let-down drum is flammable and contains CO	Let-down drum vent to fuel or flare (outside scope)	3	2	6 M	R19	Operations / Safety
N7.7	Flow	REVERSE	Back-flow from the let-down drum	Loop depressured below the drum pressure during shutdown	Flash gas or liquid back into V-201	- LV-204 fail closed - check valve (assumed)	2	1	2 L	—	—
N7.8	Integrity	OTHER THAN	Crude methanol leak	- LV-204 packing, drains, sample point - vibration of the flashing let-down line	Methanol pool fire with a nearly invisible flame; toxic exposure	- Closed drains - fire detection	4	2	8 M	R19	Operations / Safety

Node N8 · Purge PV-203, blowdown HV-205 and relief PSV-201

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N8 · Purge PV-203, blowdown HV-205 and relief PSV-201
Design intent	Hold the loop at 75 bar g (PIC-203 on PT-203) by purging 105.1 kNm ³ /h to reformer fuel through PV-203 (fail closed), keeping inerts at about 9.9 % CH ₄ ; depressure to flare through HV-205 in an emergency; relieve through PSV-201 at 88 bar g.	Design conditions	PV-203 ≈ 45 %; PSHH-203 85 bar g; PSV-201 88 bar g
Equipment	PV-203, PB-203 hand bypass, HV-205, PSV-201, flare header	Instruments	PIC-203 (PT-203), PSHH-203 (own transmitter, I-4), FI-203, AI-207
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N8.1	Pressure	MORE	Loop overpressure: purge blocked	- PV-203 fails closed (instrument air) or sticks - PIC-203 in MAN low - PB-203 closed	Pressure rises towards the make-up pressure: flange leaks, rupture if relief fails	- PIC-203 H 78 bar g - PSHH-203 I-4 at 85 bar g on its own transmitter closes XV-202 - PSV-201 88 bar g	5	1	5 M	R08, R16	Instrument · Process
N8.2	Measurement	OTHER THAN	PT-203 reads low	- Leaking impulse line - transmitter drift	PIC-203 closes the purge; the real pressure rises while the DCS shows it normal or low	- PSHH-203 I-4 on its own transmitter - PI-201 converter inlet indication	5	1	5 M	R08	Instrument
N8.3	Pressure	LESS	Loop pressure low	- PV-203 fails open - PB-203 left open - HV-205 open - make-up lost	Lower production; purge surge to the reformer fuel; circulation falls	- PIC-203 L 72 bar g - FI-203 H 150 kNm³/h	2	3	6 M	R12	Operations
N8.4	Flow	MORE	High purge	- PIC-203 SP lowered - PV-203 fails open	Carbon lost to fuel; production falls; fuel-header surge	FI-203 H 150 kNm ³ /h	2	3	6 M	—	—
N8.5	Composition	MORE	Inerts (CH ₄) build up	- Purge too small - methane slip	Lower reactant partial pressure; production falls	AI-207 H 13 %	2	3	6 M	—	—
N8.6	Flow	NO	Blowdown not available when needed	HV-205 is a manual field valve, out of reach during a fire	Loop stays at pressure in a fire: vessel rupture	PSV-201 fire case	5	2	10 H	R15	Process / Instrument
N8.7	Flow	AS WELL AS	Blowdown open in normal operation	HV-205 opened by mistake or passing	Loop gas to flare; pressure falls; flaring	- Start permissive requires HV-205 shut - PIC-203 L 72	2	2	4 L	—	—
N8.8	Safety system	OTHER THAN	Relief path blocked	PSV-201 isolation valve left closed after maintenance	No relief on overpressure	PSHH-203 I-4 at 85 bar g	5	1	5 M	R16	Process

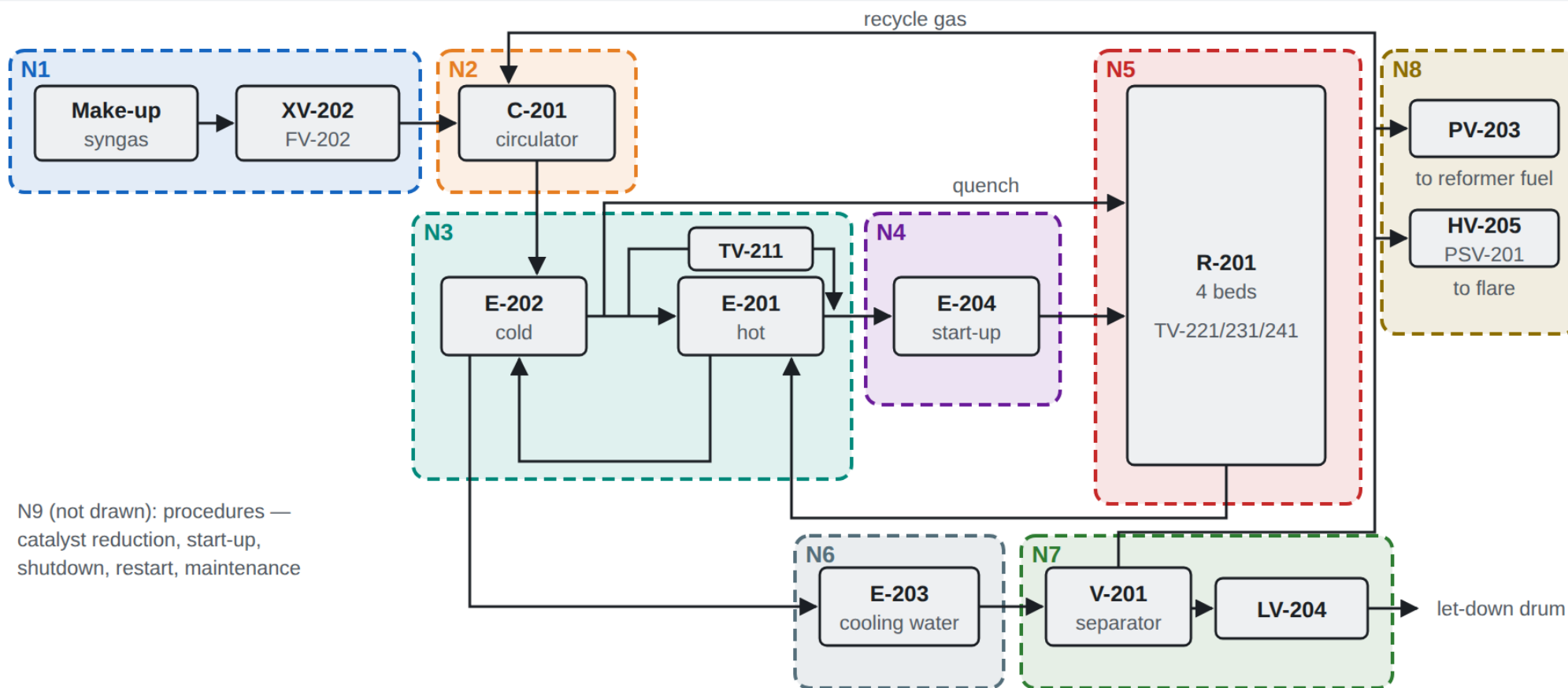
Node N9 · Procedures

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Node	N9 · Procedures: catalyst reduction, start-up and admitting make-up, shutdown, restart, maintenance
Design intent	Reduce fresh catalyst safely; heat with E-204 and admit make-up only when the SIS permissives hold; shut down with make-up off first; restart only after the cause of a trip is found; hand over for maintenance inert, isolated and passivated.	Design conditions	Procedures in the Exercises tab (cold start-up, hot start, shutdown, ESD)
Equipment	SIS sequence (reset, admit make-up), E-204, operator actions, catalyst handling	Instruments	SIS screen and permissives, HIC-214, FIC-202 (MAN), TI bed profiles
Drawing	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Document	MEOH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N9.1	Reduction	MORE	Too much H ₂ during catalyst reduction	- Wrong N₂/H₂ ratio - heating too fast	Reduction exotherm sinters the fresh catalyst (lost activity for its whole life)	- Vendor procedure - bed thermocouples	3	3	9 M	R20	Operations
N9.2	Sequence	BEFORE	Make-up admitted before the permissives hold	- Permissive bypassed - operator error	Gas meets cold or stagnant catalyst: no reaction, or a hot spot at low circulation	- Permissives: SIS reset - C-201 ≥ 50 % of design - TIC-211 and TI-212 ≥ 200 °C - no bed above 280 °C - V-201 below 70 % - HV-205 shut - FIC-202 in MAN ≤ 25 %	4	1	4 L	R22, R24	Operations
N9.3	Admission	MORE	Make-up raised too fast after admission	FIC-202 put in AUTO at the design SP at once	Sharp exotherm in bed 1 → overshoot of bed temperatures and a trip	- Permissive FIC-202 MAN ≤ 25 % - TI alarms - TSHH I-1	3	3	9 M	R22	Operations
N9.4	Sequence	AFTER	Make-up shut off after the circulator is stopped (shutdown)	Wrong order in a planned shutdown	Make-up meets stagnant hot catalyst: hot spot	FSLL-201 I-2 closes XV-202	4	1	4 L	R23	Operations
N9.5	Restart	EARLY	Restart straight after a trip without finding the cause	Pressure to restore production	Same upset repeats, second trip from a hotter bed	- SIS reset blocked while a trip condition is present - permissives	3	3	9 M	R28	Operations
N9.6	Maintenance	OTHER THAN	Isolation incomplete for vessel entry	- N₂ or loop gas passing isolation valves - blinds missing	Asphyxiation or CO exposure inside the vessel	- Permit to work - gas testing	5	1	5 M	R21, R29	Operations / Safety · Process
N9.7	Unloading	OTHER THAN	Reduced catalyst exposed to air	Converter opened or unloaded without passivation	Pyrophoric catalyst glows: burns, fire of wax residues, CO fumes	Passivation step in the vendor procedure	4	2	8 M	R21	Operations / Safety
N9.8	Safety system	OTHER THAN	Trip bypass left in after testing	- Bypass not removed - no DCS indication	Unsafe condition goes unprotected	Administrative control	4	2	8 M	R24	Operations
N9.9	Sampling	OTHER THAN	CO exposure at sample points	- Manual sampling of make-up, loop gas or crude - analyzer vents	CO poisoning; H ₂ fire in the analyzer house	- Sampling permit - ventilation	4	2	8 M	R19	Operations / Safety

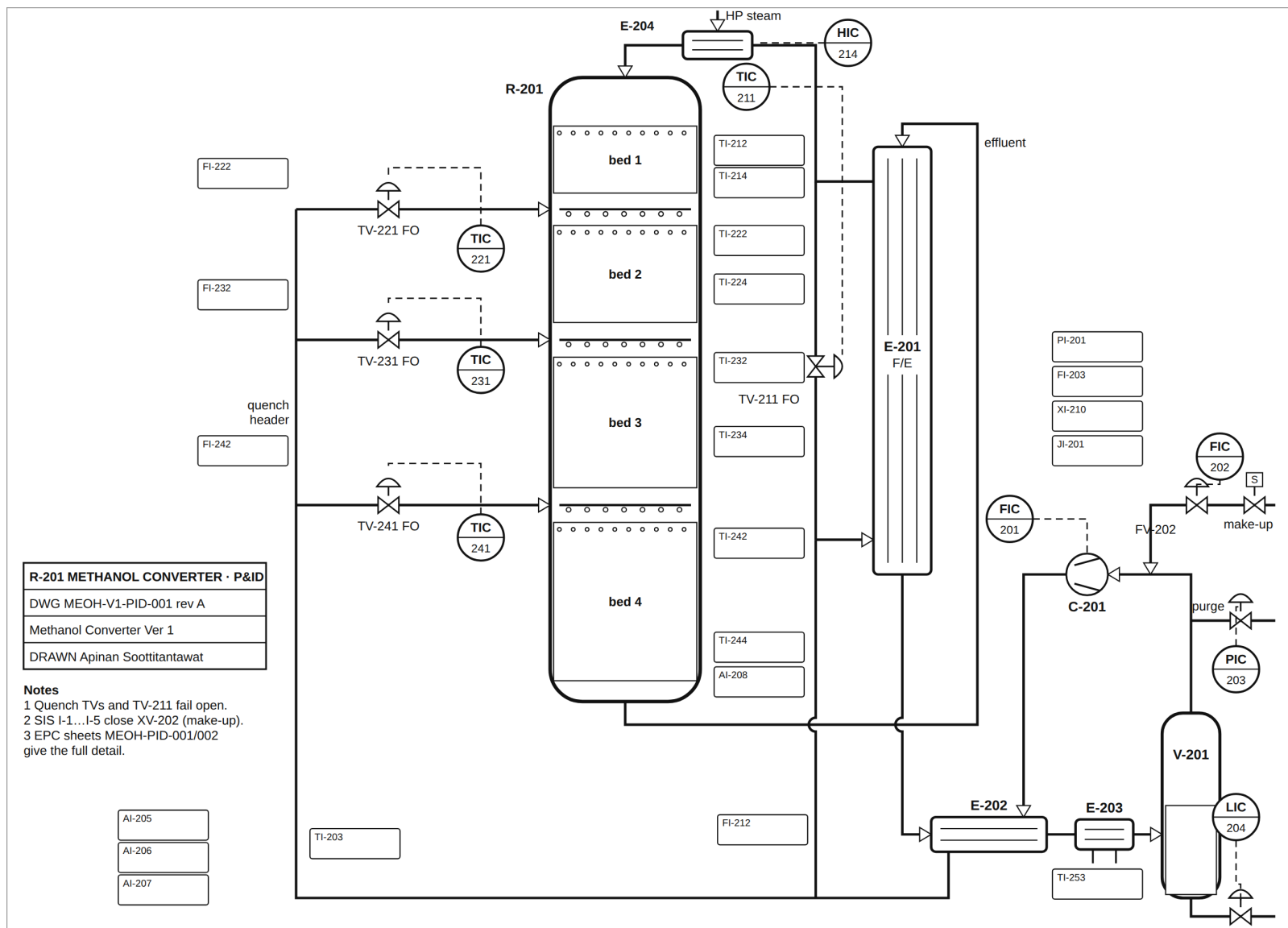
Appendix 2 · HAZOP documents (drawings)

A2-1 · HAZOP node map



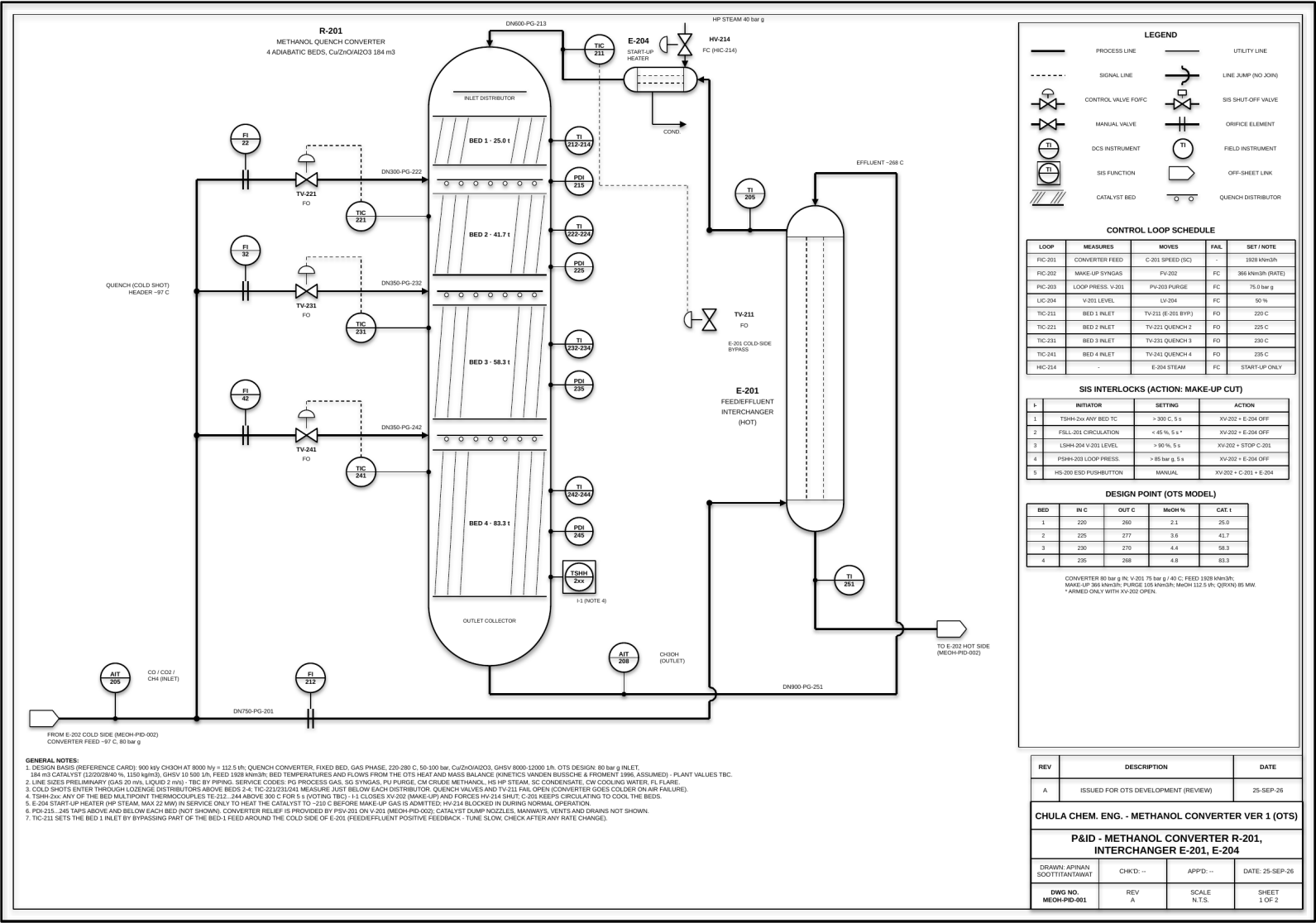
A2-1. HAZOP node map of the R-201 loop (nodes N1-N8; N9 = procedures).

A2-2 · Simulator P&ID (screen drawing)



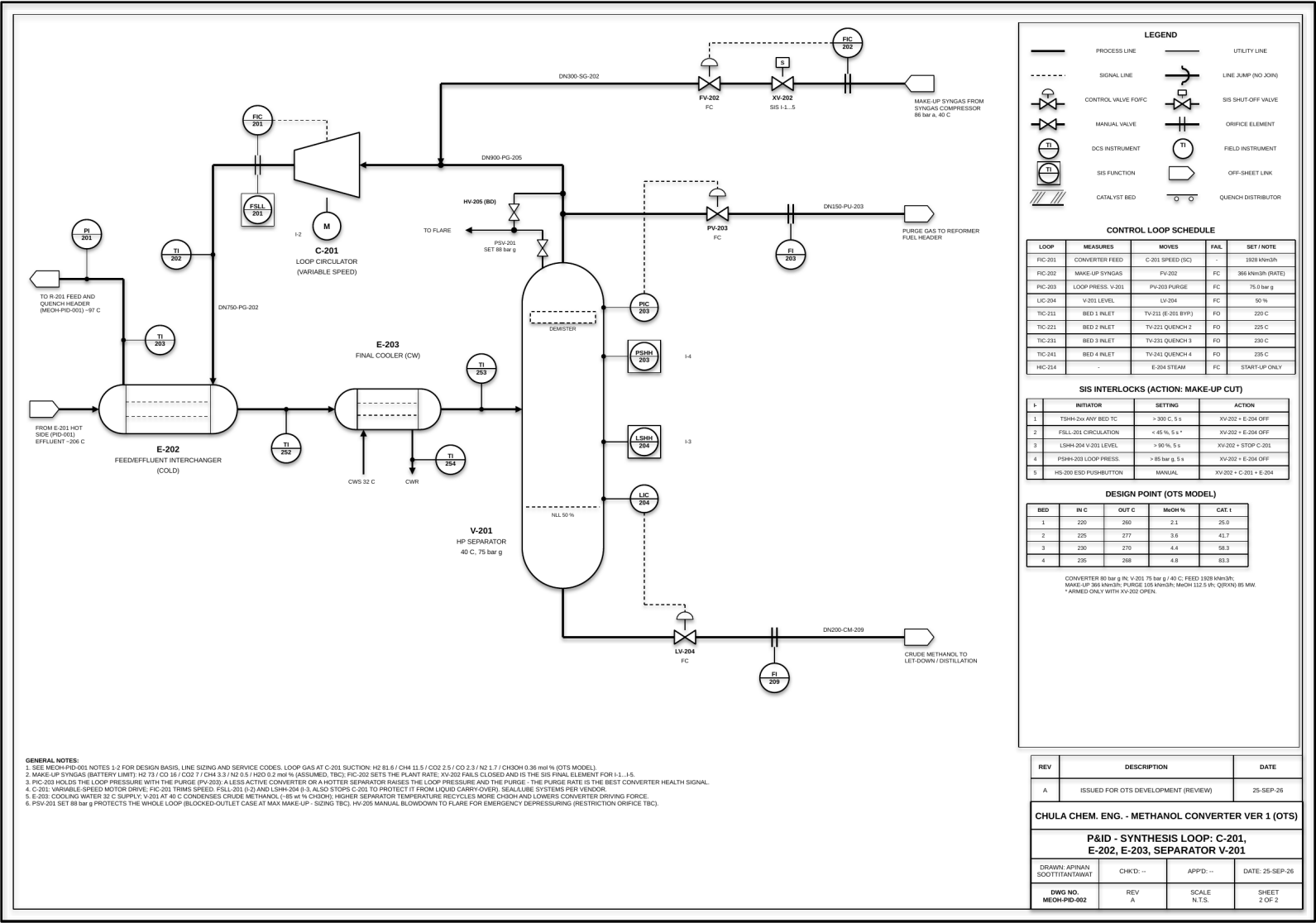
A2-2. The P&ID of the simulator (P&ID tab), the reference for the console, the exercises and this study.

A2-3 · MEOH-PID-001 (R-201, E-201, E-204)



A2-3. EPC P&ID sheet 1 of 2: R-201 converter, interchanger E-201 and start-up heater E-204.

A2-4 · MEOH-PID-002 (synthesis loop)



A2-4. EPC P&ID sheet 2 of 2: C-201, E-202, E-203, separator V-201, purge and let-down.

Appendix 3 · HAZOP action close-out sheets

One sheet per recommendation (29 sheets). The action party writes the response, ticks the status and signs; the HAZOP leader accepts the close-out.

HAZOP ACTION CLOSE-OUT SHEET · R01

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R01
Node	N2 · Circulator C-201 and recycle gas	Action party	Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.1	Flow NO : No circulation	• C-201 motor trip • power failure • LSHH-204 or ESD stops C-201	If make-up continues: gas meets stagnant hot catalyst → hot spot in bed 1, sintering	• FSLL-201 I-2 below 45 % (868 kNm³/h) closes XV-202 — uses FT-201 of FIC-201 • FIC-201 L 1700 kNm³/h	S4 L2 8 Medium
N2.2	Flow LESS : Low circulation	• FIC-201 SP lowered • fouled impeller (performance loss) • FT-201 reading high	Less heat carried out of the beds per kmol → bed exits hotter; at low enough flow the TSHH trips	• FIC-201 L 1700 • TI-224/234/244 H 285 °C • TSHH I-1 • FSLL-201 I-2	S4 L2 8 Medium
N2.4	Measurement OTHER THAN : FT-201 fails reading high	Impulse-line or transmitter fault	FIC-201 cuts the real circulation and the FSLL-201 trip reads the same wrong value: loss of circulation not seen by I-2	• Bed temperature alarms • TSHH I-1 (independent of flow)	S4 L2 8 Medium

HAZOP recommendation

FT-201B + FSLL-201B: own flow transmitter on the C-201 discharge for the low-circulation trip, so I-2 no longer shares FT-201 with FIC-201.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R02

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R02
Node	N7 · Separator V-201 and crude methanol let-down LV-204	Action party	Instrument / Process
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.1	Level LESS : Low level: gas blow-by through LV-204	• LV-204 fails open • LB-204 bypass left open • LT-204 frozen or reading high • LIC-204 in MAN with a high output	The boot empties (section 9: in a few minutes with LV-204 wide open) and loop gas at 75 bar g flows into the let-down drum at about 6 bar g: drum overpressure and rupture, large release, fire and explosion	• LIC-204 L 25 % • LV-204 fail closed	S5 L2 10 High
N7.2	Measurement OTHER THAN : LT-204 freezes	Transmitter fault	LIC-204 holds a constant output while the real level drifts: overfill (LSHH-204 protects) or empty (nothing independent protects)	LSHH-204 I-3 on its own transmitter (high side only)	S5 L2 10 High

HAZOP recommendation

LT-204B + LSLL-204 + XV-204: low-low level on its own transmitter closing a new let-down shut-off valve (gas blow-by protection); LOPA to set its SIL.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R03

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R03
Node	N7 · Separator V-201 and crude methanol let-down LV-204	Action party	Process
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.1	Level LESS : Low level: gas blow-by through LV-204	• LV-204 fails open • LB-204 bypass left open • LT-204 frozen or reading high • LIC-204 in MAN with a high output	The boot empties (section 9: in a few minutes with LV-204 wide open) and loop gas at 75 bar g flows into the let-down drum at about 6 bar g: drum overpressure and rupture, large release, fire and explosion	• LIC-204 L 25 % • LV-204 fail closed	S5 L2 10 High

HAZOP recommendation

PSV-205 on the let-down drum sized for full gas blow-by through LV-204 (and LB-204) at loop pressure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R04

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R04
Node	N5 · Converter R-201	Action party	Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Flow MORE : More make-up than the loop can convert	- FIC-202 SP raised (production push) - FT-202 reads low - FV-202 opened in MAN	More heat released in beds 1 and 2 → bed exits towards the catalyst limit; more purge and inerts	- FIC-202 H 400 kNm³/h - TI-214 H 275 °C - TI-224/234/244 H 285 °C - TSHH I-1 at 300 °C closes XV-202	S4 L2 8 Medium
N1.4	Composition OTHER THAN : CO-rich make-up	Reformer or shift upset: CO up, CO ₂ down	Much more heat per kmol converted in beds 1 and 2: bed exits approach the trip; sintering; by-products	- TIC-211/221/231/241 - TI-214 H 275 °C - TI-224 H 285 °C - TSHH I-1 at 300 °C	S4 L2 8 Medium
N2.2	Flow LESS : Low circulation	- FIC-201 SP lowered - fouled impeller (performance loss) - FT-201 reading high	Less heat carried out of the beds per kmol → bed exits hotter; at low enough flow the TSHH trips	- FIC-201 L 1700 - TI-224/234/244 H 285 °C - TSHH I-1 - FSL-201 I-2	S4 L2 8 Medium
N3.2	Temperature MORE : Bed 1 inlet too hot	- TV-211 sticks closed - TIC-211 thermocouple reading low - E-204 steam passing	Bed 1 exit rises; catalyst sintering; runaway risk	- TIC-211 H 228 °C - TI-214 H 275 °C - TSHH I-1 at 300 °C	S4 L2 8 Medium
N5.1	Flow NO : No quench to bed 2	- TV-221 fails closed or sticks - TIC-221 in MAN at 0 % - TIC-221 thermocouple reads high	Bed 2 inlet rises by tens of °C; bed 2 exit approaches the trip; sintering; runaway	- TIC-221 H 232 °C - TI-224 H 285 °C - TSHH I-1 at 300 °C	S4 L2 8 Medium
N5.3	Temperature MORE : Bed temperature runaway	- CO-rich gas - loss of circulation with make-up on - high make-up - E-204 steam passing	Catalyst above ~300 °C: sintering, by-products (wax, higher alcohols, methane); hot gas at the shell → shell overheating	- TI-214 H 275 °C, TI-224/234/244 H 285 °C - TSHH I-1 at 300 °C (same elements as the indication) - FSL-201 I-2 - XV-202 closes	S4 L2 8 Medium
N5.6	Flow PART OF : Channelling or poor quench mixing	- Damaged quench distributor - catalyst settling	Part of the gas bypasses the catalyst; local hot and cold zones that one thermocouple misses	Three thermocouples per bed (upper third, lower third, exit)	S3 L2 6 Medium
N5.7	Measurement OTHER THAN : Bed thermocouple fails low	Open circuit (reads ambient)	That element cannot trip the TSHH; the operator loses the bed-exit reading	- TI-212/213 in the same bed - the other TSHH elements	S4 L2 8 Medium

HAZOP recommendation

TSHH 2003: three elements per bed exit (TE-214A/B/C ...) voted 2003, independent of the indication thermocouples.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R05

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R05
Node	N1 · Make-up gas	Action party	Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.8	Pressure MORE : Loop pressure driven up by the make-up	- Loop not consuming gas (reaction stopped) while make-up flows - front-end compressor at its maximum	Loop pressure rises towards the make-up pressure 86 bar g: flange leaks, rupture if relief fails	- PIC-203 H 78 bar g - PSHH-203 I-4 at 85 bar g closes XV-202 - PSV-201 88 bar g	S5 L1 5 Medium
N1.9	Flow REVERSE : Loop gas back into the front end	Front-end pressure falls below the loop with FV-202 open and XV-202 still open	Loop depressures backwards; surge of the front-end compressor	- XV-202 closes only on a loop trip - check valve on the make-up line (assumed)	S3 L2 6 Medium
N1.10	Safety system OTHER THAN : XV-202 fails to close on demand	Stuck valve, solenoid failure: one final element serves every trip	Every trip that relies on stopping the make-up fails: runaway, overpressure or blow-by continue	- FIC-202 is also driven to MAN 0 % by the trip (FV-202 closes) - proof testing	S5 L1 5 Medium
N5.3	Temperature MORE : Bed temperature runaway	- CO-rich gas - loss of circulation with make-up on - high make-up - E-204 steam passing	Catalyst above ~300 °C: sintering, by-products (wax, higher alcohols, methane); hot gas at the shell → shell overheating	- TI-214 H 275 °C, TI-224/234/244 H 285 °C - TSHH I-1 at 300 °C (same elements as the indication) - FSL-201 I-2 - XV-202 closes	S4 L2 8 Medium

HAZOP recommendation

XV-202B: second make-up shut-off valve in series (1002) so one stuck valve does not defeat every trip; proof-test both.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R06

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R06
Node	N5 · Converter R-201	Action party	Instrument / Mechanical
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.3	Temperature MORE : Bed temperature runaway	• CO-rich gas • loss of circulation with make-up on • high make-up • E-204 steam passing	Catalyst above ~300 °C: sintering, by-products (wax, higher alcohols, methane); hot gas at the shell → shell overheating	• TI-214 H 275 °C, TI-224/234/244 H 285 °C • TSHH I-1 at 300 °C (same elements as the indication) • FSLL-201 I-2 • XV-202 closes	S4 L2 8 Medium
N5.9	Temperature MORE : Hot gas at the pressure shell	• Internal leak into the annulus • insulation damage	Shell above its design temperature: HTHA, creep, rupture	None on the P&ID (no skin thermocouples)	S5 L1 5 Medium

HAZOP recommendation

TAHH-2xx wall: converter shell skin thermocouples with alarm (hot gas reaching the pressure shell).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R07

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R07
Node	N3/N5 · E-202 cold and E-201 hot interchangers with the TV-211 bypass / Converter R-201	Action party	Operations / Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.4	Integrity OTHER THAN: E-201 internal leak	• Tube or tubesheet failure • cold feed at the higher pressure	Feed bypasses the converter: lower conversion and effluent temperature, hard to diagnose	• AI-208 L 3.8 % • XI-210 L 105 t/h	S2 L3 6 Medium
N5.8	Composition OTHER THAN: Catalyst ageing or poisoning	• Sintering over the years • sulfur or chloride slip	Lower activity; inlets raised to compensate; hot zone moves down	• Bed profiles • AI-208 L 3.8 %	S2 L4 8 Medium

HAZOP recommendation

AIT-208 validation: lab check of converter-outlet methanol each shift; alarm on analyzer frozen (no change for a set time).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R08

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R08
Node	N8 · Purge PV-203, blowdown HV-205 and relief PSV-201	Action party	Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N8.1	Pressure MORE : Loop overpressure: purge blocked	- PV-203 fails closed (instrument air) or sticks - PIC-203 in MAN low - PB-203 closed	Pressure rises towards the make-up pressure: flange leaks, rupture if relief fails	- PIC-203 H 78 bar g - PSHH-203 I-4 at 85 bar g on its own transmitter closes XV-202 - PSV-201 88 bar g	S5 L1 5 Medium
N8.2	Measurement OTHER THAN : PT-203 reads low	- Leaking impulse line - transmitter drift	PIC-203 closes the purge; the real pressure rises while the DCS shows it normal or low	- PSHH-203 I-4 on its own transmitter - PI-201 converter inlet indication	S5 L1 5 Medium

HAZOP recommendation

Deviation alarm between PT-203 (PIC-203) and the PSHH-203 transmitter, so a low-reading PT-203 is seen before the trip point.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R09

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R09
Node	N2 · Circulator C-201 and recycle gas	Action party	Mechanical
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.7	Flow REVERSE : Reverse flow through C-201 after a stop	C-201 stops while the discharge side is at higher pressure	Machine turns backwards; seal and bearing damage; leak	Discharge non-return valve (vendor, assumed)	S3 L2 6 Medium

HAZOP recommendation

Confirm a non-return valve on the C-201 discharge and the vendor anti-surge / reverse-rotation protection.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R10

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R10
Node	N1 · Make-up gas	Action party	Instrument / Operations
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.4	Composition OTHER THAN : CO-rich make-up	Reformer or shift upset: CO up, CO ₂ down	Much more heat per kmol converted in beds 1 and 2: bed exits approach the trip; sintering; by-products	- TIC-211/221/231/241 - TI-214 H 275 °C - TI-224 H 285 °C - TSHH I-1 at 300 °C	S4 L2 8 Medium
N1.5	Composition OTHER THAN : CO ₂ -rich make-up	Reformer upset: CO down, CO ₂ up	Less reaction heat, more water in the crude, production falls	- AI-206 indication - XI-210 L 105 t/h	S2 L3 6 Medium
N1.7	Composition AS WELL AS : Poisons or metal carbonyls	- Sulfur or chloride slip from the front end - Fe/Ni carbonyls from carbon steel in cold CO service	Top of bed 1 loses activity; heat release moves down; faster ageing; wax and hot spots from Fe/Ni deposits	- Front-end guard beds (outside scope) - bed-1 profile TI-212/213/214	S3 L3 9 Medium

HAZOP recommendation

Make-up composition analyzer (CO, CO₂) with alarm and a written response to a reformer upset (cut FIC-202, raise quench margins).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R11

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R11
Node	N6 · E-203 final cooler (cooling water)	Action party	Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.6	Temperature MORE : Hot suction gas	- Cooling-water loss at E-203 - hot afternoon	Higher discharge temperature and power; methanol vapour recycled to the converter	- TI-253 H 45 °C - TI-202 indication	S2 L3 6 Medium
N6.1	Flow LESS : Less cooling water	- CW pump trip - valve throttled - fouling	Separator hot: methanol stays in the gas and is recycled; production falls; purge and loop pressure rise; hotter C-201 suction	- TI-253 H 45 °C - PIC-203 H 78 - FI-203 H 150 - XI-210 L 105 t/h	S3 L3 9 Medium
N6.2	Flow NO : No cooling water	- CW header lost - both valves closed	As above, faster; CW trapped in the tubes may boil	TI-253 H 45 °C	S3 L2 6 Medium

HAZOP recommendation

Low-flow alarm on the cooling water to E-203 (TI-253 alone reacts late).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R12

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R12
Node	N7/N8 · Separator V-201 and crude methanol let-down LV-204 / Purge PV-203, blowdown HV-205 and relief PSV-201	Action party	Operations
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.1	Level LESS : Low level: gas blow-by through LV-204	• LV-204 fails open • LB-204 bypass left open • LT-204 frozen or reading high • LIC-204 in MAN with a high output	The boot empties (section 9: in a few minutes with LV-204 wide open) and loop gas at 75 bar g flows into the let-down drum at about 6 bar g: drum overpressure and rupture, large release, fire and explosion	• LIC-204 L 25 % • LV-204 fail closed	S5 L2 10 High
N8.3	Pressure LESS : Loop pressure low	• PV-203 fails open • PB-203 left open • HV-205 open • make-up lost	Lower production; purge surge to the reformer fuel; circulation falls	• PIC-203 L 72 bar g • FI-203 H 150 kNm³/h	S2 L3 6 Medium

HAZOP recommendation

Lock LB-204 and PB-203 bypass valves closed; bypass use only under permit with an operator stationed at the valve.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R13

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R13
Node	N6 · E-203 final cooler (cooling water)	Action party	Process
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.4	Integrity OTHER THAN : Tube leak: loop gas into the cooling water	- Corrosion - vibration	Loop gas carried to the cooling tower: flammable and toxic gas at the tower; CW overpressure	CW-side relief (assumed)	S4 L2 8 Medium
N6.6	Pressure MORE : CW side blocked in with hot gas flowing	CW inlet and outlet valves closed	Boiling and overpressure of the CW side	CW-side relief (assumed)	S3 L2 6 Medium

HAZOP recommendation

Gas detection or high-pressure alarm on the E-203 cooling-water return; confirm CW-side relief for tube rupture.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R14

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R14
Node	N4 · E-204 start-up steam heater	Action party	Process / Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.1	Flow MORE : Steam in normal operation (valve passing)	- HIC-214 valve leaking through - block valve left open	Bed 1 inlet hotter; TV-211 opens to compensate and hides it; wasted steam; with TV-211 saturated bed 1 overheats	- TIC-211 H 228 °C - FI-214 indication - TI-214 H 275 °C	S2 L3 6 Medium
N4.4	Integrity OTHER THAN : E-204 tube leak (gas into steam)	Tube failure	Loop gas at 80 bar g enters the steam and condensate system: overpressure, flammable gas at condensate vents	Steam block valve closed in normal operation	S4 L2 8 Medium

HAZOP recommendation

Double block and bleed on E-204 steam when idle; steam-side relief sized for tube rupture; steam-flow alarm when HIC-214 is at 0 %.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R15

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R15
Node	N8 · Purge PV-203, blowdown HV-205 and relief PSV-201	Action party	Process / Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.8	Integrity OTHER THAN: Seal or casing leak	• Dry-gas seal failure • vibration • liquid slug	Loop gas release at 75 bar g in the compressor house: jet or flash fire, CO exposure	• Seal-gas monitoring • ESD I-5 stops C-201 and closes XV-202 • HV-205	S4 L2 8 Medium
N3.5	Integrity OTHER THAN: External leak of hot gas at E-201 or the converter nozzles	• Thermal cycling at start-up and trips • bolt relaxation • gasket failure	Jet fire of hot H ₂ /CO/methanol vapour at up to 268 °C with a nearly invisible flame; CO exposure	• Gas detection (if fitted) • ESD • HV-205 blowdown (manual)	S5 L2 10 High
N8.6	Flow NO: Blowdown not available when needed	HV-205 is a manual field valve, out of reach during a fire	Loop stays at pressure in a fire: vessel rupture	PSV-201 fire case	S5 L2 10 High

HAZOP recommendation

Make HV-205 a remotely operated blowdown valve (fail open on ESD or a separate button) with a depressuring rate per API 521 for fire.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R16

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R16
Node	N8 · Purge PV-203, blowdown HV-205 and relief PSV-201	Action party	Process
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.8	Pressure MORE : Loop pressure driven up by the make-up	• Loop not consuming gas (reaction stopped) while make-up flows • front-end compressor at its maximum	Loop pressure rises towards the make-up pressure 86 bar g: flange leaks, rupture if relief fails	• PIC-203 H 78 bar g • PSHH-203 I-4 at 85 bar g closes XV-202 • PSV-201 88 bar g	S5 L1 5 Medium
N3.6	Pressure MORE : Exchanger side blocked in while hot	Isolation for maintenance with heat on the other side	Thermal expansion of trapped gas → overpressure of the isolated side	PSV-201 covers only equipment connected to the loop	S3 L1 3 Low
N8.1	Pressure MORE : Loop overpressure: purge blocked	• PV-203 fails closed (instrument air) or sticks • PIC-203 in MAN low • PB-203 closed	Pressure rises towards the make-up pressure: flange leaks, rupture if relief fails	• PIC-203 H 78 bar g • PSHH-203 I-4 at 85 bar g on its own transmitter closes XV-202 • PSV-201 88 bar g	S5 L1 5 Medium
N8.8	Safety system OTHER THAN : Relief path blocked	PSV-201 isolation valve left closed after maintenance	No relief on overpressure	PSHH-203 I-4 at 85 bar g	S5 L1 5 Medium

HAZOP recommendation

Confirm PSV-201 relief cases (fire, blocked purge with make-up at maximum pressure, E-201/E-204 tube rupture) and car-seal the relief isolation valves.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R17

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R17
Node	N3 · E-202 cold and E-201 hot interchangers with the TV-211 bypass	Action party	Mechanical / Safety
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.5	Integrity OTHER THAN: External leak of hot gas at E-201 or the converter nozzles	• Thermal cycling at start-up and trips • bolt relaxation • gasket failure	Jet fire of hot H ₂ /CO/methanol vapour at up to 268 °C with a nearly invisible flame; CO exposure	• Gas detection (if fitted) • ESD • HV-205 blowdown (manual)	S5 L2 10 High

HAZOP recommendation

Flange management for hot H₂ service (controlled bolting, hot bolt-up); H₂/CO gas detection and UV/IR flame detection at E-201 and the converter nozzles.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R18

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R18
Node	N3/N5 · E-202 cold and E-201 hot interchangers with the TV-211 bypass / Converter R-201	Action party	Materials
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.8	Material OTHER THAN: Hot-section steel below its hydrogen limit (HTHA)	- Material not per API 941 at H₂ partial pressure ≈ 65 bar a - missing PWHT	Hidden hydrogen attack → sudden rupture, large release, fire	- Material selection - inspection	S5 L1 5 Medium
N5.9	Temperature MORE: Hot gas at the pressure shell	- Internal leak into the annulus - insulation damage	Shell above its design temperature: HTHA, creep, rupture	None on the P&ID (no skin thermocouples)	S5 L1 5 Medium

HAZOP recommendation

Verify hot-section materials against the current API 941 (Nelson curves) at the trip temperature plus margin; HTHA inspection programme.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R19

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R19
Node	N7/N9 · Separator V-201 and crude methanol let-down LV-204 / Procedures	Action party	Operations / Safety
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N7.6	Composition AS WELL AS : Dissolved gas in the crude	H ₂ , CO, CO ₂ dissolved at 75 bar g	Flash gas at the let-down drum is flammable and contains CO	Let-down drum vent to fuel or flare (outside scope)	S3 L2 6 Medium
N7.8	Integrity OTHER THAN : Crude methanol leak	• LV-204 packing, drains, sample point • vibration of the flashing let-down line	Methanol pool fire with a nearly invisible flame; toxic exposure	• Closed drains • fire detection	S4 L2 8 Medium
N9.9	Sampling OTHER THAN : CO exposure at sample points	• Manual sampling of make-up, loop gas or crude • analyzer vents	CO poisoning; H ₂ fire in the analyzer house	• Sampling permit • ventilation	S4 L2 8 Medium

HAZOP recommendation

Closed-loop sampling for make-up, loop gas and crude methanol; CO detectors in the analyzer house; personal CO monitors.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R20

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R20
Node	N9 · Procedures	Action party	Operations
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.3	Temperature MORE : Too much heating at start-up	- HIC-214 opened quickly - make-up admitted with a hot bed	Beds above target when make-up arrives → sharp exotherm and trip	- Permissive: no bed above 280 °C - TI alarms - TSHH I-1	S3 L2 6 Medium
N9.1	Reduction MORE : Too much H ₂ during catalyst reduction	- Wrong N₂/H₂ ratio - heating too fast	Reduction exotherm sinters the fresh catalyst (lost activity for its whole life)	- Vendor procedure - bed thermocouples	S3 L3 9 Medium

HAZOP recommendation

Catalyst reduction procedure with hold points (H₂ in N₂ steps, bed ΔT limit, water make) and simulator briefing before each reduction.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R21

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R21
Node	N9 · Procedures	Action party	Operations / Safety
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.6	Maintenance OTHER THAN : Isolation incomplete for vessel entry	• N₂ or loop gas passing isolation valves • blinds missing	Asphyxiation or CO exposure inside the vessel	• Permit to work • gas testing	S5 L1 5 Medium
N9.7	Unloading OTHER THAN : Reduced catalyst exposed to air	Converter opened or unloaded without passivation	Pyrophoric catalyst glows: burns, fire of wax residues, CO fumes	Passivation step in the vendor procedure	S4 L2 8 Medium

HAZOP recommendation

Catalyst unloading and entry procedure: passivation end-point or unloading under N₂; inert-entry rules (supplied air, standby, rescue plan); positive isolation.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R22

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R22
Node	N9 · Procedures	Action party	Operations
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.2	Flow MORE : More make-up than the loop can convert	- FIC-202 SP raised (production push) - FT-202 reads low - FV-202 opened in MAN	More heat released in beds 1 and 2 → bed exits towards the catalyst limit; more purge and inerts	- FIC-202 H 400 kNm³/h - TI-214 H 275 °C - TI-224/234/244 H 285 °C - TSHH I-1 at 300 °C closes XV-202	S4 L2 8 Medium
N4.3	Temperature MORE : Too much heating at start-up	- HIC-214 opened quickly - make-up admitted with a hot bed	Beds above target when make-up arrives → sharp exotherm and trip	- Permissive: no bed above 280 °C - TI alarms - TSHH I-1	S3 L2 6 Medium
N4.5	Sequence LATE : Steam left on after the reaction is self-sustaining	HIC-214 or its block valve not closed (the SIS closes the steam only on a trip)	As steam passing; E-201 duty falls, loop runs hotter at the same TIC-211 output	- SIS closes E-204 steam on every trip - procedure	S2 L3 6 Medium
N9.2	Sequence BEFORE : Make-up admitted before the permissives hold	- Permissive bypassed - operator error	Gas meets cold or stagnant catalyst: no reaction, or a hot spot at low circulation	- Permissives: SIS reset - C-201 ≥ 50 % of design - TIC-211 and TI-212 ≥ 200 °C - no bed above 280 °C - V-201 below 70 % - HV-205 shut - FIC-202 in MAN ≤ 25 %	S4 L1 4 Low
N9.3	Admission MORE : Make-up raised too fast after admission	FIC-202 put in AUTO at the design SP at once	Sharp exotherm in bed 1 → overshoot of bed temperatures and a trip	- Permissive FIC-202 MAN ≤ 25 % - TI alarms - TSHH I-1	S3 L3 9 Medium

HAZOP recommendation

Start-up procedure: admit make-up in MAN at ≤ 25 %, raise slowly while watching bed 1-2 exits; close E-204 steam and its block valve once self-sustaining; train on the simulator.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R23

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R23
Node	N9 · Procedures	Action party	Operations
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.4	Sequence AFTER : Make-up shut off after the circulator is stopped (shutdown)	Wrong order in a planned shutdown	Make-up meets stagnant hot catalyst: hot spot	FSSL-201 I-2 closes XV-202	S4 L1 4 Low

HAZOP recommendation

Shutdown procedure: make-up off first, then reduce circulation; never stop C-201 with make-up on.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R24

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R24
Node	N9 · Procedures	Action party	Operations
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.2	Sequence BEFORE : Make-up admitted before the permissives hold	• Permissive bypassed • operator error	Gas meets cold or stagnant catalyst: no reaction, or a hot spot at low circulation	• Permissives: SIS reset • C-201 $\geq 50\%$ of design • TIC-211 and TI-212 $\geq 200\text{ }^{\circ}\text{C}$ • no bed above $280\text{ }^{\circ}\text{C}$ • V-201 below 70% • HV-205 shut • FIC-202 in MAN $\leq 25\%$	S4 L1 4 Low
N9.8	Safety system OTHER THAN : Trip bypass left in after testing	• Bypass not removed • no DCS indication	Unsafe condition goes unprotected	Administrative control	S4 L2 8 Medium

HAZOP recommendation

SIS bypass management: permit, time limit, DCS indication of every bypass, removal checked at shift handover.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R25

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R25
Node	N2/N7 · Circulator C-201 and recycle gas / Separator V-201 and crude methanol let-down LV-204	Action party	Instrument / Mechanical
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.5	Composition AS WELL AS : Liquid in the circulator suction	- V-201 high level - demister damaged - separator hot	Liquid slugs damage impeller and seals → gas leak; circulation lost	- LIC-204 H 75 % - LSHH-204 I-3 at 90 % stops C-201 and closes XV-202 - TI-253 H 45 °C	S3 L2 6 Medium
N2.8	Integrity OTHER THAN : Seal or casing leak	- Dry-gas seal failure - vibration - liquid slug	Loop gas release at 75 bar g in the compressor house: jet or flash fire, CO exposure	- Seal-gas monitoring - ESD I-5 stops C-201 and closes XV-202 - HV-205	S4 L2 8 Medium
N7.3	Level MORE : High level in V-201	- LV-204 sticks closed - LT-204 frozen - let-down drum blocked	Liquid carried into the C-201 suction: impeller and seal damage, gas leak, circulation lost	- LIC-204 H 75 % - LSHH-204 I-3 at 90 % on its own transmitter closes XV-202 and stops C-201	S3 L2 6 Medium
N7.5	Composition AS WELL AS : Liquid carried over with the gas	- Demister damaged - high gas velocity	Methanol recycled to the converter (equilibrium penalty); liquid to C-201	None direct	S2 L3 6 Medium

HAZOP recommendation

Demister ΔP indication on V-201 and a vibration trip on C-201 (liquid carry-over).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R26

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R26
Node	N5 · Converter R-201	Action party	Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Temperature LESS : Bed 1 inlet too cold	• TV-211 fails open (instrument air) or sticks open • E-201 fouled • converter outlet cooler	Bed 1 slows, the heat front moves down, the reaction can die out; production falls	• TIC-211 L 212 °C • XI-210 L 105 t/h	S2 L3 6 Medium
N3.2	Temperature MORE : Bed 1 inlet too hot	• TV-211 sticks closed • TIC-211 thermocouple reading low • E-204 steam passing	Bed 1 exit rises; catalyst sintering; runaway risk	• TIC-211 H 228 °C • TI-214 H 275 °C • TSHH I-1 at 300 °C	S4 L2 8 Medium
N5.1	Flow NO : No quench to bed 2	• TV-221 fails closed or sticks • TIC-221 in MAN at 0 % • TIC-221 thermocouple reads high	Bed 2 inlet rises by tens of °C; bed 2 exit approaches the trip; sintering; runaway	• TIC-221 H 232 °C • TI-224 H 285 °C • TSHH I-1 at 300 °C	S4 L2 8 Medium
N5.2	Flow MORE : Too much quench (valve fails open)	• TV-231 fails open • instrument air failure (FO) • TIC thermocouple reads low	Next bed too cold; reaction shifts to the other beds, bed 1 and 2 run hotter as the main-feed share falls; production falls	• TIC-231 L 223 °C • TI-214 H 275 °C • XI-210 L 105 t/h	S3 L3 9 Medium

HAZOP recommendation

Position feedback (ZT) on TV-211 and TV-221/231/241 with an output-vs-position deviation alarm (stuck or failed quench valve).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R27

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R27
Node	N1 · Make-up gas	Action party	Instrument
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.11	Utility OTHER THAN: Instrument air failure	Air compressor trip, header break	All valves to fail positions: FV-202, PV-203, LV-204 shut; TV-211 and quench valves open → reaction dies, level rises slowly (see section 9)	• Fail positions (FC/FO) chosen for a safe state • XV-202 de-energise-to-trip	S2 L3 6 Medium

HAZOP recommendation

Verify on site that every valve goes to its fail position on instrument-air loss; keep XV-202 de-energise-to-trip.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R28

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R28
Node	N9 · Procedures	Action party	Operations
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No make-up gas	- Front-end syngas compressor trip - XV-202 closed by a trip or spurious closure - FV-202 fails closed (instrument air)	Production stops; the loop keeps circulating and the beds cool until the reaction dies; loop pressure falls as the purge continues; restart needs the permissives	- FIC-202 L 330 kNm³/h - XI-210 L 105 t/h - PIC-203 closes the purge	S2 L3 6 Medium
N5.4	Temperature LESS : Beds too cold, reaction dies	- Make-up lost - quench open - E-201 fouled	Production stops; restart needs reheating through E-204	- XI-210 L 105 t/h - TIC L alarms	S2 L3 6 Medium
N9.5	Restart EARLY : Restart straight after a trip without finding the cause	Pressure to restore production	Same upset repeats, second trip from a hotter bed	- SIS reset blocked while a trip condition is present - permissives	S3 L3 9 Medium

HAZOP recommendation

Restart procedure after a trip: find the cause, reset only when the trip condition has cleared, reheat through E-204, then admit make-up per R22.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R29

Project	Methanol Converter Ver 1 · R-201 loop Operator Training Simulator	Document	MEOH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R29
Node	N9 · Procedures	Action party	Process
Drawings	MEOH-PID-001 (R-201, E-201, E-204) and MEOH-PID-002 (C-201, E-202, E-203, V-201, purge and let-down)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N9.6	Maintenance OTHER THAN : Isolation incomplete for vessel entry	• N₂ or loop gas passing isolation valves • blinds missing	Asphyxiation or CO exposure inside the vessel	• Permit to work • gas testing	S5 L1 5 Medium

HAZOP recommendation

Nitrogen utility connections by removable spool or double block and bleed with check valve (no syngas backflow into the N₂ header).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			